MIL-STD-188-171
19 MAY 1989
SUPERSEDING
SEE FOREWORD

MILITARY STANDARD

INTEROPERABILITY STANDARDS FOR INFORMATION AND RECORD TRAFFIC EXCHANGE MODE I



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FOREWORD

1. This military standard is approved and mandatory for use by all Departments and Agencies of the Department of Defense in accordance with Department of Defense Directive Number 4640.11, December 21, 1987 (appendix A).

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Joint Tactical Command, Control, and Communications Agency
ATTN: C3A-ADW-S
11440 Isaac Newton Square North
Reston, Virginia 22090-5006

by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. Originally, Military Standard 188 (MIL-STD-188) covered technical standards for tactical and long haul communications, but later evolved through revisions (MIL-STD-188A, MIL-STD-188B) into a document applicable to tactical communications only (MIL-STD-188C).

4. The Defense Communications Agency (DCA) published DCA circulars (DCAC) promulgating standards and engineering criteria applicable to the long haul Defense Communications System (DCS) and to the technical support of the National Military Command System (NMCS). The Joint Tactical Communications Office (JTCO) published specifications promulgating standards applicable to the joint tactical communication system known as the TRI-TAC system.

5. As a result of a Joint Chiefs of Staff (JCS) action, standards for all military communications are now being published in a MIL-STD-188 series of documents. The MIL-STD-188 series is subdivided into a MIL-STD-188-100 series covering common standards for tactical and long haul communications, a MIL-STD-188-200 series covering standards for tactical communications only, and a MIL-STD-188-300 series covering standards for long haul communications only. Emphasis is being placed on developing common standards for tactical and long haul communications published in the MIL-STD-188-100 series.

6. This document is the result of JCS action requiring that the technical characteristics of channel coordination procedures, previously contained in various specifications and DCA circulars, be updated and published in the MIL-STD-188 series of standards. This document contains the technical requirements for Mode I. The current MIL-STD-188 series for information and record traffic exchange is as follows:

a.	MIL-STD-188-171	Mode I
b.	MIL-STD-188-172	Mode II
c.	MIL-STD-188-173	Mode V
d.	MIL-STD-188-174	Mode VI
e.	MIL-STD-TBD	Mode VII

7. Mode I was previously described in DCAC 370-D175-1.

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1. SCOPE

1.1 Purpose. The purpose of this document is to ensure interoperability and to promote commonality of communications equipment and subsystems using Mode I channel coordination procedures. An additional objective is to prevent proliferation of equipment serving the same or similar functions. The variety of equipment shall be the minimum necessary to support the missions of the armed forces in accordance with Department of Defense Directive 4630.5, Compatibility and Interoperability of Tactical Command, Control, Communications, and Intelligence Systems.

1.2 Scope. This document specifies the minimum requirements necessary to implement information and record traffic exchange among data communications systems employed in long-haul and tactical communications networks. It is not the intent of this document to specify any particular hardware or software design or implementation. Message formats are not specified in this document.

1.3 Application. This document is applicable to the design and development of new equipment, assemblages, and systems. Existing systems that are undergoing major modification or rehabilitation shall comply with the standards contained in this document, subject to the applicable requirements of current procurement regulations. This document shall not be used solely to justify retrofit of existing systems. This document, in whole or in part, is applicable only to the extent specified in the procurement documents of an equipment item or system.

1.4 System standards and design objectives. The parameters and other requirements specified in this document are mandatory system standards if the word "shall" is used in connection with the parameter or requirement under consideration. Nonmandatory system standards and design objectives are identified as optional by the word "should" in connection with the parameter or requirement under consideration. For a definition of the terms "system standards" and "design objectives", see FED-STD-1037. Information paragraphs, shown as notes, have been included to better define certain methods currently in use.

1.5 Tailoring. As a minimum, those features or functions specified herein necessary to ensure interoperability among systems shall be implemented in an equipment item. While every effort has been made to include all the features necessary for protocol implementation, certain aspects are dependent upon system application and must be tailored by the specification writer. These aspects include alarm functions, block size, data rates, codes, message formats, message size, message numbering, etc.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the current issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto.

STANDARDS

FEDERAL

FED-STD-1037	Glossary of Telecommunication Terms
FIPS PUB 17-1	Character Structure and Character Parity Sense for Serial-By-Bit Data Communications in the Code for Information Interchange

MILITARY

MIL-STD-188-100	Common Long Haul and Tactical Communication Systems Technical Standards

2.2 Order of precedence. In the event of a conflict between this document and the references cited herein, the text of this document takes precedence. Nothing in this standard, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.3 Source of documents. Copies of the referenced federal standards are available from:

Department of Defense
Single Stock Point
Commanding Officer
Naval Publications and Forms Center
5801 Tabor Avenue
Philadelphia, PA 19120-5099

For specific acquisition functions, these documents should be obtained from the contracting activity or as directed by the contracting activity.

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3. DEFINITIONS

3.1 Definitions of terms. Definitions of terms used in this document are as specified in FED-STD-1037A. Additional terms, unique to information and record traffic exchange, are defined below.

3.1.1 Character set. The character set used in Mode I consists of 256 octets and is in compliance with the eight-bit environment of FIPS PUB 17-1. The octets are divided into two 128-character sets, one set containing an odd number of logical 1 bits per octet and the other set containing an even number of logical 1 bits per octet. The octet set containing the odd number of 1 bits maps directly into the eight-bit American Standard Code for Information Interchange (ASCII) character set, i.e., the odd parity eight-bit code. The set of octets used for control and framing has an even number of logical 1 bits (an all-logical 0 octet is considered to have an even number of logical 1 bits).

3.1.2 Data rate. The rate, in bits per second (bps), at which Mode I octets are transferred between terminals (transmitter and receiver).

3.1.3 Mode I. Mode I is an automatic repeat-request (ARQ) channel coordination procedure that provides for synchronous, simultaneous, duplex data transfer. It is designed to be used over terrestrial links. Messages are structured into blocks. Blocks are transmitted to the receiver. The receiver positively acknowledges error-free blocks and negatively acknowledges blocks with errors. All blocks that are negatively acknowledged are retransmitted. A block of data cannot be transmitted until the previous block has been positively acknowledged. (In block-by-block operation, no portion of the next block shall be transmitted until the block has been acknowledged. In continuous operation, the next block may be transmitted up to, but not including, the ETX/ETB characters.)

3.1.4 Mode II. Mode II is a non-ARQ channel coordination procedure which provides for asynchronous, simultaneous, independent, duplex data transfer. No acknowledgment procedure is implemented.

3.1.5 Mode V. Mode V is an ARQ channel coordination procedure which provides for asynchronous, simultaneous, duplex data transfer. It is designed to be used over terrestrial links. Error-free messages are positively acknowledged. Messages with errors are rejected by the receiver, and retransmission is requested.

3.1.6 Mode VI. Mode VI is an ARQ channel coordination procedure which provides for synchronous, simultaneous, duplex data transfer. It is designed to be used over terrestrial and satellite links. Messages are structured into blocks. Each block has a unique number associated with it. Error-free blocks are positively acknowledged. Blocks containing errors are negatively acknowledged. All acknowledgments, positive and negative, have the unique block number associated with them. The blocks of a message are organized into block groups. All blocks in a block group are transmitted whether or not the preceding blocks have been positively acknowledged. Any blocks not positively acknowledged are retransmitted. All blocks in a block group must be positively acknowledged before transmission of the next block group is initiated.

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3.1.7 Mode VII. Mode VII is an ARQ channel coordination procedure which provides for synchronous, simultaneous, duplex data transfer. It is designed to be used over terrestrial and satellite links. Messages are structured into frames. Frames without errors are acknowledged. All frames received with errors are retransmitted.

3.2 Definitions of abbreviations and acronyms. Abbreviations and acronyms used in this document are as defined in FED-STD-1037A and are provided in appendix B.

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4. GENERAL REQUIREMENTS

4.1 Mode I. The Mode I channel coordination procedure provides for synchronous, simultaneous, duplex data transfer. It is designed to be used over terrestrial links and includes features to ensure message integrity. Mode I also uses a partition scheme to distinguish control and data characters.

4.2 Duplex operation. Duplex operation shall be achieved by providing both transmitter and receiver functions at each end of a transmission link. These transmitters and receivers shall be capable of simultaneously sending and receiving messages.

4.3 Octet code. The requirement to be able to transmit the entire set of 128 ASCII characters (see table I) necessitated development of an eight-bit (octet) coded control system that is transparent to all data characters. The Mode I character set shall consist of 256 eight-bit characters (see table II). The octet structure shall not have a parity bit. The Mode I code shall be divided into two 128 eight-bit character groups. One group shall consist of characters containing an even number of logical 1 bits. This group shall be used for control and framing. The other group shall consist of characters containing an odd number of logical 1 bits. This group shall be used for data. Characters (data, control, and framing) shall be transmitted serial-by-bit, serial-by-character (least significant bit first) in compliance with FIPS PUB 17-1. The eight-bit Mode I code shall comply with the FIPS PUB 17-1 eight-bit coded environment character structure.

4.4 Characters. Mode I operation shall use three generic character types: data, control, and framing. These characters are identified in the bit stream by their sequence and position relative to other characters. Table II defines the Mode I character set. The generic types are described below:

4.4.1 Data characters. There are 128 data characters as defined in table II. For convenience, these characters are also shown in table I; data characters are 8 bits long and have an odd number of logical 1 bits.

4.4.2 Framing and control characters. Framing and control characters have an even number of logical 1 bits except for Block Parity (BP). BP may be any one of the 256 octets. For convenience, control and framing characters are also shown in table III.

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TABLE I. American Standard Code for Information Interchange (ASCII) and Mode I data characters.

Bits a_4	a_3	a_2	a_1	Row \ Column →	0 (0 0 0)	1 (0 0 1)	2 (0 1 0)	3 (0 1 1)	4 (1 0 0)	5 (1 0 1)	6 (1 1 0)	7 (1 1 1)
0	0	0	0	0	NUL	DLE	SP	0	@	P	`	p
0	0	0	1	1	SOH	DC1	!	1	A	Q	a	q
0	0	1	0	2	STX	DC2	"	2	B	R	b	r
0	0	1	1	3	ETX	DC3	#	3	C	S	c	s
0	1	0	0	4	EOT	DC4	$	4	D	T	d	t
0	1	0	1	5	ENQ	NAK	%	5	E	U	e	u
0	1	1	0	6	ACK	SYN	&	6	F	V	f	v
0	1	1	1	7	BEL	ETB	' (APOS)	7	G	W	g	w
1	0	0	0	8	BS	CAN	(	8	H	X	h	x
1	0	0	1	9	HT	EM	)	9	I	Y	i	y
1	0	1	0	10	LF	SUB	*	:	J	Z	j	z
1	0	1	1	11	VT	ESC	+	;	K	[	k	{
1	1	0	0	12	FF	FS	,	<	L	\	l	|
1	1	0	1	13	CR	GS	-	=	M	]	m	}
1	1	1	0	14	SO	RS	.	>	N	^	n	~
1	1	1	1	15	SI	US	/	?	O	_	o	DEL

Note: Bit position a_8 (not shown) shall be generated such that the character shall have an odd number of logical 1 bits.
a_1 = LSB; a_8 = MSB

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TABLE II. Mode I data, control, and framing 8-bit environment set.

Bits a_8 a_7 a_6 a_5 →	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
a_4 a_3 a_2 a_1 ↓ / Row ↓ / Column →	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0 0 0 0	0		DLE	SP		@	P_F		p	NUL			Ø		P		
0 0 0 1	1	SOH	REP_C		1	A_F	Q	a		SOH_F	DC1	!		A			q
0 0 1 0	2	STX	RM_C		2	B_F	R	b		STX_F	DC2	"		B			r
0 0 1 1	3	ETX_F	DC3	#		C	S_F		s	ETX			3	C_F	S	c	
0 1 0 0	4	EOT			4	D_F	T	d			DC4	$		D			t
0 1 0 1	5	NOTE 6	NAK	%		E			u	ENQ	NAK_C		5	E_F	U	e	
0 1 1 0	6	$ACK1_C$	SYN	&		F			v	ACK	SYN_C		6	F_F	V	f	
0 1 1 1	7	BEL	ETB_F		7	G_F	W	g		INV_C	ETB	(APOS)		G			w
1 0 0 0	8	BS	CAN_C		8	H_F	X	h			CAN	(		H			x
1 0 0 1	9		EM	)		I			y	HT	EM_C		9		Y	i	
1 0 1 0	10		SUB	*		J			z	LF	MC_C		:	J_F	Z	j	
1 0 1 1	11				;	K_F	[	k		VT$_C$	ESC	+		K			{
1 1 0 0	12		FS	,		L			:	FF	$ACK2_C$		<		\	l	
1 1 0 1	13	CR			=	M_F	]	m		NOTE 6	GS	-		M			}
1 1 1 0	14	SO	WRJ_C		>		^	n		NOTE 6	RS	.		N			~
1 1 1 1	15	NOTE 6	US	/		0			DEL	SI			?		_	o	DEL_F

NOTES

1 ■ = not used at this time, available for future use

2 Subscript C = control character

3 Subscript F = frame character

4 a_1 = LSB, a_8 = MSB

5 Framing characters E, G, J, K, M, and P are reserved for future use.

6 These characters are reserved for special internal equipment use and shall not be transmitted over the loop

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TABLE III. Mode I eight bit control and framing codes.

Bits a_4	a_3	a_2	a_1	Row	Column 0 (000)	Column 1 (001)	Column 2 (010)	Column 3 (011)	Column 4 (100)	Column 5 (101)	Column 6 (110)	Column 7 (111)
0	0	0	0	0						P		
0	0	0	1	1	SOH	REP			A			
0	0	1	0	2	STX	RM			B			
0	0	1	1	3	ETX				C	S		
0	1	0	0	4					D			
0	1	0	1	5	NOTE 6	NAK			E			
0	1	1	0	6	ACK1	SYN			F			
0	1	1	1	7	INV	ETB			G			
1	0	0	0	8		CAN			H			
1	0	0	1	9	NOTE 6	EM						
1	0	1	0	10		MC			J			
1	0	1	1	11					K			
1	1	0	0	12		ACK2						
1	1	0	1	13	NOTE 6				M			
1	1	1	0	14	NOTE 6	WBT						
1	1	1	1	15	NOTE 6							DEL

NOTES:
1. ■ = not used at this time; available for future use.
2. Bit position a_8 (not shown) shall be generated such that the character has an even number of logical 1 bits.
3. ACK 1, ACK 2, NAK, INV, REP, RM, WBT, and CAN are transmitted in identical contiguous pairs.
4. a_1 = LSB; a_8 = MSB
5. Framing characters E, G, J, K, M, and P are reserved for future use.
6. These characters are reserved for special internal equipment use and shall not be transmitted over the loop.

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4.4.2.1 Framing characters. Framing characters serve to delineate the beginning and end of each block of data. Two framing characters precede the block and two succeed the block (see figures 1 through 11). The framing characters are:

a. Start of header (SOH) - the first framing character of the first block of a message,

b. Select characters (A, B, C, D, E, F, G, H, J, K, M, P, S) (see tables III and IV) - follows SOH.

c. Start of text (STX) - the first framing character of all blocks after the first block of the message.

d. Delete character (DEL) - always follows STX.

e. End of transmission block (ETB) - the third framing character of all message blocks except the last.

f. End of text (ETX) - the third framing character of the last block of a message.

g. Block parity (BP) - the last framing character at the end of all blocks.

4.4.2.2 Control characters. The control characters are:

a. Acknowledgment number one (ACK1)

b. Acknowledgment number two (ACK2)

c. Negative acknowledgment (NAK)

d. Reject message (RM)

e. Wait before transmit (WBT)

f. Reply (REP)

g. Cancel (CAN)

h. End of medium (EM)

i. Invalid sequence (INV)

j. Mode change (MC)

NOTE: DC-3 may be both a data and a special control character (see 5.6.2.4).

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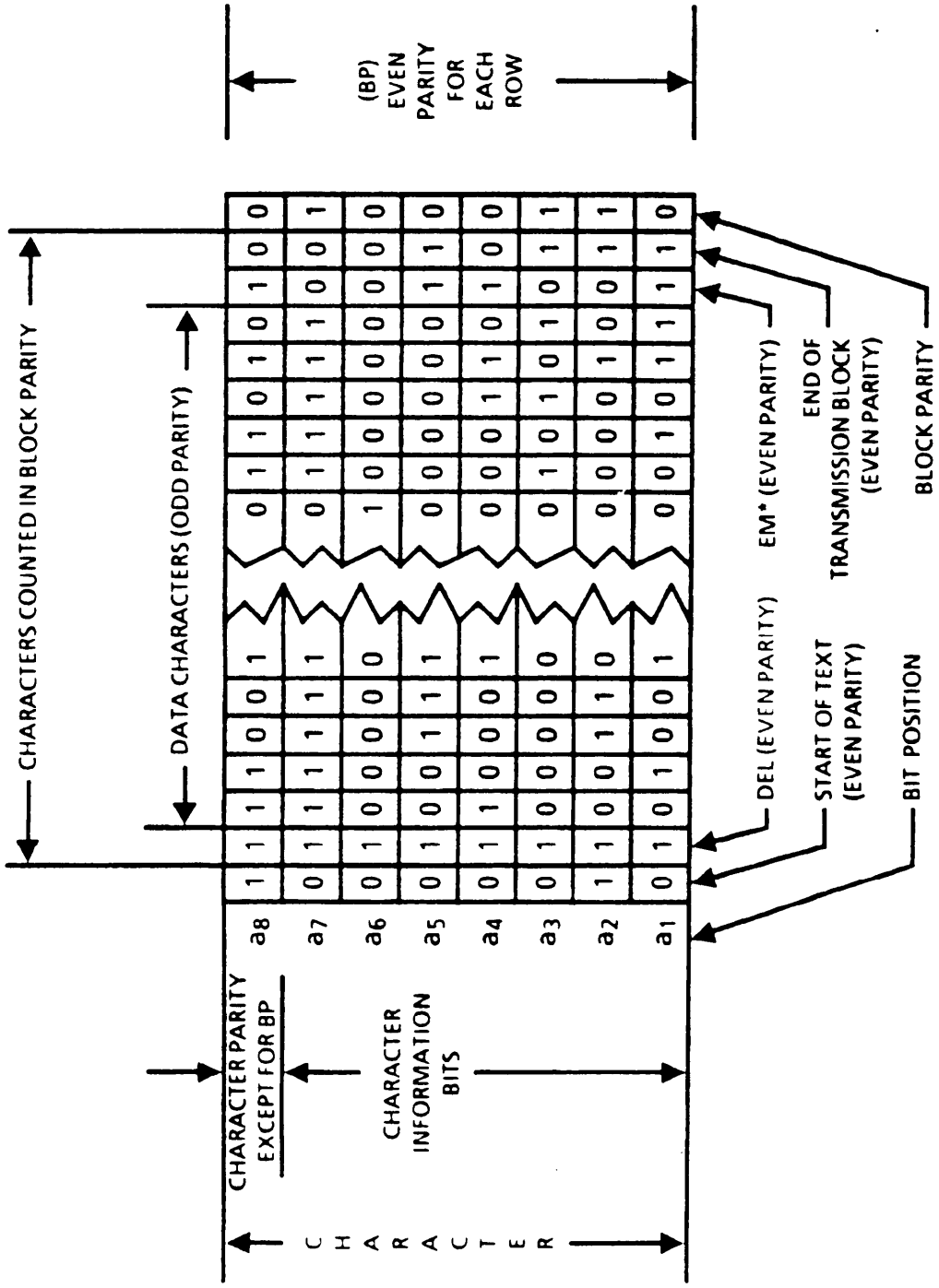


FIGURE 1. Mode I parity checking code rules for Mode I message blocks.

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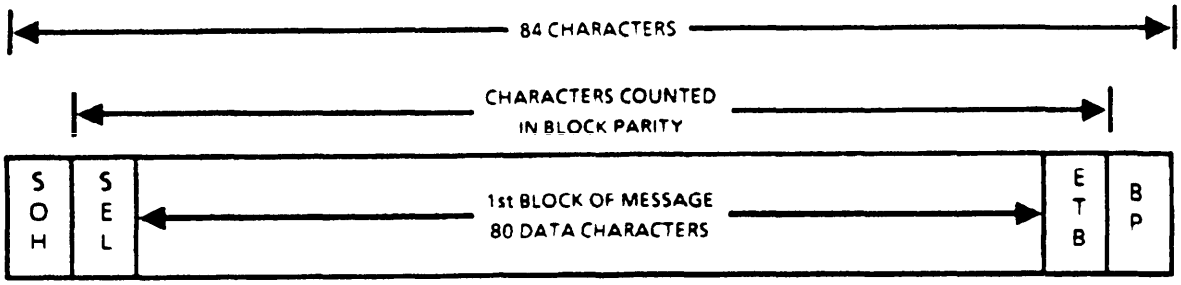


FIGURE 2. Mode I header block - normal block - 1st block of message.

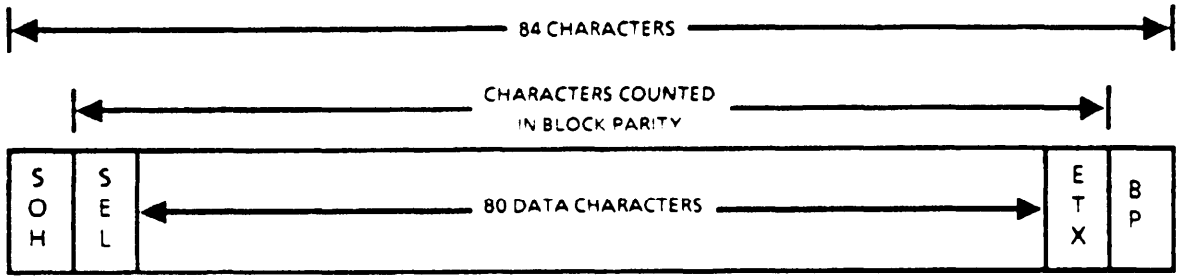


FIGURE 3. Mode I header block - normal block - one block message.

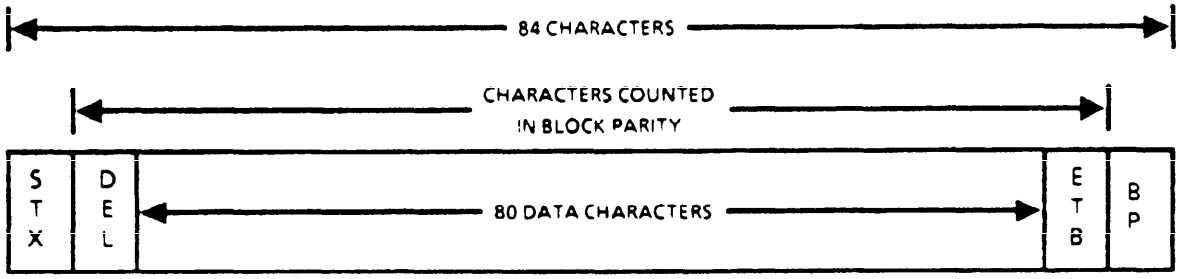


FIGURE 4. Mode I text block - normal block.

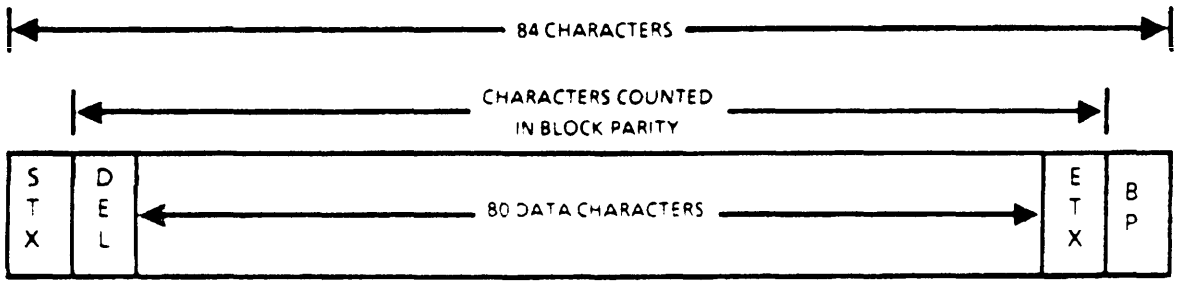


FIGURE 5. Mode I text block - normal block - last block of message.

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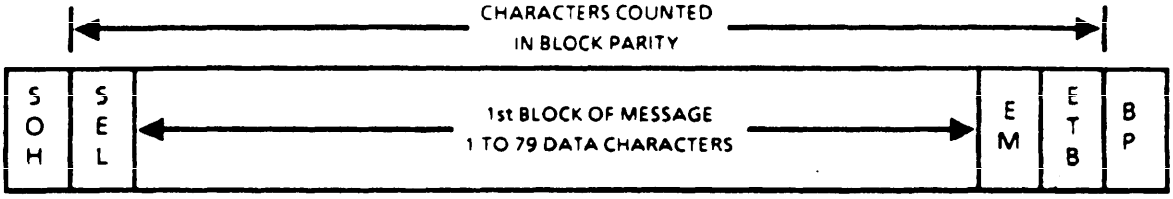


FIGURE 6. Mode I header block - short block - 1st block of message.

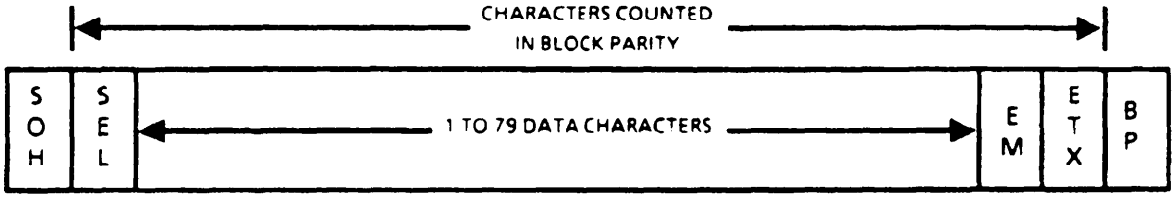


FIGURE 7. Mode I header block - short block - one block message.

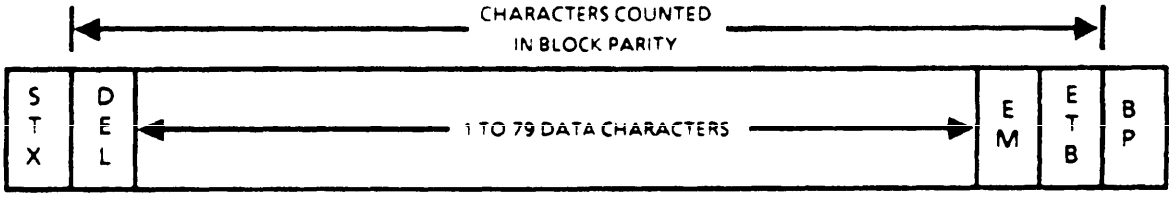


FIGURE 8. Mode I text block - short block.

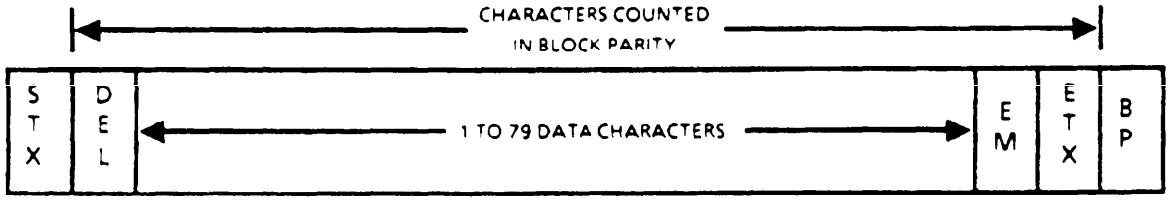


FIGURE 9. Mode I text block - short block - last block of message.

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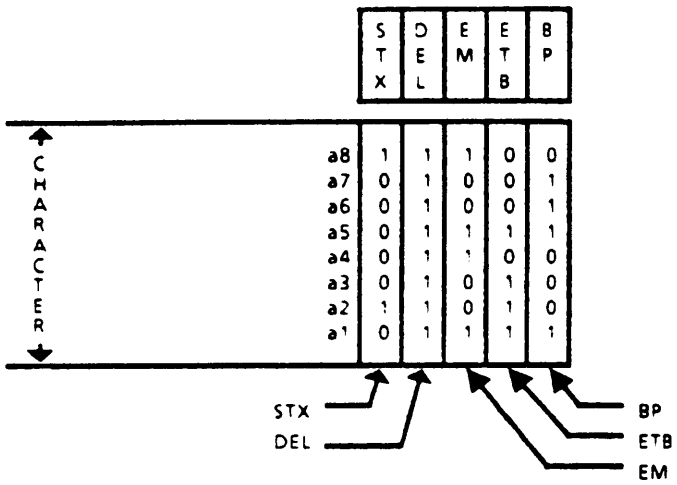


FIGURE 10. Mode I example of EM block.

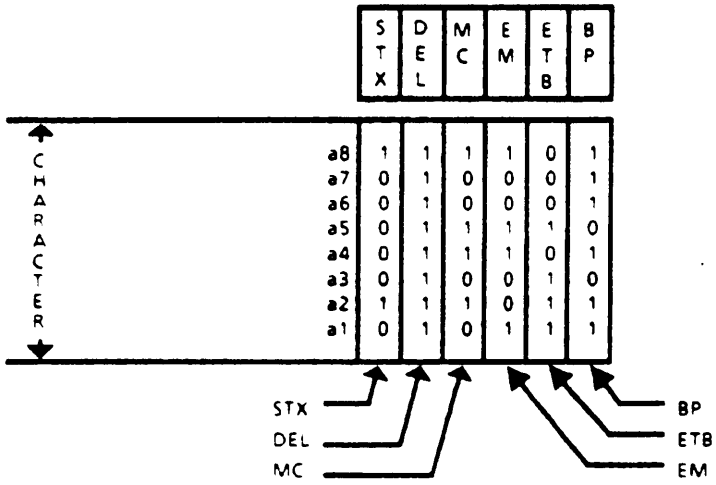


FIGURE 11. Mode I example of MC block beginning/end of binary text.

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Table IV. Mode I select framing characters.

Select Character	Code								Identification
	a_8	a_7	a_6	a_5	a_4	a_3	a_2	a_1	
A	0	1	0	0	0	0	0	1	IA No. 2 (Note 1)
B	0	1	0	0	0	0	1	0	Identifies the source transmission media as being 7 track. (Note 1)
C	1	1	0	0	0	0	1	1	Identifies the source transmission media as being 9 track. (Note 1)
D	0	1	0	0	0	1	0	0	Card format (Note 1)
E	1	1	0	0	0	1	0	1	Reserved for future use.
F	1	1	0	0	0	1	1	0	Card format (Note 2) Activate operator alarm.
G	0	1	0	0	0	1	1	1	Reserved for future use.
H	0	1	0	0	1	0	0	0	Mode I data code. (Note 1)
J	1	1	0	0	1	0	1	0	Reserved for future use.
K	0	1	0	0	1	0	1	1	Reserved for future use.
M	0	1	0	0	1	1	0	1	Reserved for future use.
P	0	1	0	1	0	0	0	0	Reserved for future use.
S	0	1	0	1	0	0	1	1	Activate operator alarm. (Note 3)

NOTES:

1 Select Characters A, B, C, D, and H are used by the receiver to route messages to the output device associated with the Select Character.

2 Select Character F will be inserted by the message switch as the second framing character of the first block of each card message delivered to a Mode I channel if the header of the message indicates flash or higher precedence. It shall not be inserted by any subscriber terminal.

3 Select Character S will be inserted by the message switch as the second framing character of the first block of each paper tape message delivered to a Mode I channel if the header of the message indicates flash or higher precedence. Select Character S shall not be generated by any terminal

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4.5 Message integrity. Messages shall be structured, by the transmitter, into blocks of up to 80 data characters. Message integrity shall be accomplished by positive acknowledgment of error-free blocks; i.e., blocks with proper framing characters, correct character parity, and valid block parity. Control characters, as discussed in subsequent sections, shall accomplish the message transmission and acknowledgment processes.

4.6 Message format. The message format to be used shall be agreed upon by the system managers.

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5. DETAILED REQUIREMENTS

5.1 Mode I channel coordination procedure. This section specifies an information exchange procedure for synchronous transmission systems. Mode I features include:

a. Duplex data exchange capabilities,

b. Automatic acknowledgment of receipt of message blocks,

c. Error detection with automatic repeat-request,

d. Message integrity,

e. Channel synchronization, and

f. Safe message store.

5.2 Block structure. Each message block shall consist of no more than 80 data characters and four framing characters. A properly framed block shall be a block in which all of the framing characters are correctly positioned. Correct positions for the framing characters are depicted in figures 1 through 11. A character sequence that does not exhibit proper block framing shall be ignored. All received blocks that are properly framed shall be acknowledged.

5.3 Message structure. Mode I messages shall be segmented into blocks. A message shall contain one or more blocks. Messages shall consist of normal blocks, short blocks, or combinations of the two.

5.3.1 Normal message blocks. Normal blocks shall contain 80 data characters and 4 framing characters (see figures 2, 3, 4, and 5).

5.3.2 Short message blocks. Short message blocks shall consist of from 0 to 79 data characters (see figures 6, 7, 8, and 9), four framing characters, and an end of medium (EM) control character. The total length of the short message block may vary from five to 84 characters. Short blocks may be used at any time and may be intermixed with normal blocks. The criteria for transmission of short blocks shall be specified in the applicable procurement document. All Mode I systems shall be capable of accepting a short block of any length.

NOTE: Some current systems are not capable of accepting short blocks and will be classmarked as such.

5.3.3 Data codes. Mode I shall support the data codes described below.

5.3.3.1 Mode I character set. Mode I shall support the Mode I data character set (see table I).

5.3.3.2 ASCII. Mode I shall support the ASCII character set with odd parity. This ASCII character set correlates with the Mode I data character set.

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5.3.3.3 Other data exchange codes and binary bit streams. Mode I shall be capable of supporting other data exchange codes and binary bit streams. Such codes and bit streams shall be segmented into seven-bit data elements with an eighth bit added to create an octet with an odd number of logical 1 bits (see appendix C).

5.4 Block-by-block and continuous operation. There shall be two Mode I procedures: block-by-block operation and continuous operation. In block-by-block operation, the next block is not transmitted until the last block transmitted has been positively acknowledged (ACK). In continuous operation, transmission of the next block can start prior to receipt of the acknowledgment for the last block transmitted (see figures 12 through 16 and appendix D).

5.4.1 Block-by-block operation. Block-by-block operation shall be characterized by the transmission of blocks such that each block must be positively acknowledged (ACK) before the transmission of the next block. If the expected ACK to a block is received (see 5.6.1.1.1), then the next block shall be transmitted. If the wrong ACK is received, then the message shall be canceled and retransmitted. If a negative acknowledgment (NAK) is received, the block shall be retransmitted. If no answer has been received at the expiration of the answer timer, the reply (REP) sequence shall be sent. The answer to the REP sequence shall determine the next operation. Figures 12 and 13 are examples of block-by-block operation.

5.4.2 Continuous operation. Continuous operation shall be characterized by beginning the transmission of the next block before receiving the acknowledgment of the last block transmitted. However, the end of block sequence of the next block shall not be transmitted until an ACK has been received for the last block transmitted. When the 82nd character or end of medium (EM) character of the next block is sent and an answer for the last block transmitted has not been received, synchronous idle (SYN) characters shall be transmitted until an answer is received (see figures 14 and 15). If an answer is expected, the following shall apply:

a. If the answer is received and it is the expected ACK sequence, then the next block shall be completed with end of block framing sequence, end of transmission block/block parity (ETB/BP), or end of text/block parity (ETX/BP).

b. If the answer to a block is the wrong ACK, then the entire message shall be canceled and retransmitted.

c. If, while the next block is being transmitted, a NAK sequence for the last block transmitted is received, then the transmitter shall cease transmission of the next block at a character boundary and immediately retransmit the block that caused transmission of the NAK.

d. If the answer timer (see 5.9.1) expires during the transmission of a block, then the transmitter shall wait until the 82nd character or EM character of that block is sent before sending the REP sequence.

e. If the block transmitted is the last block of the message, then further transmission shall be suspended until the expected ACK sequence is received.

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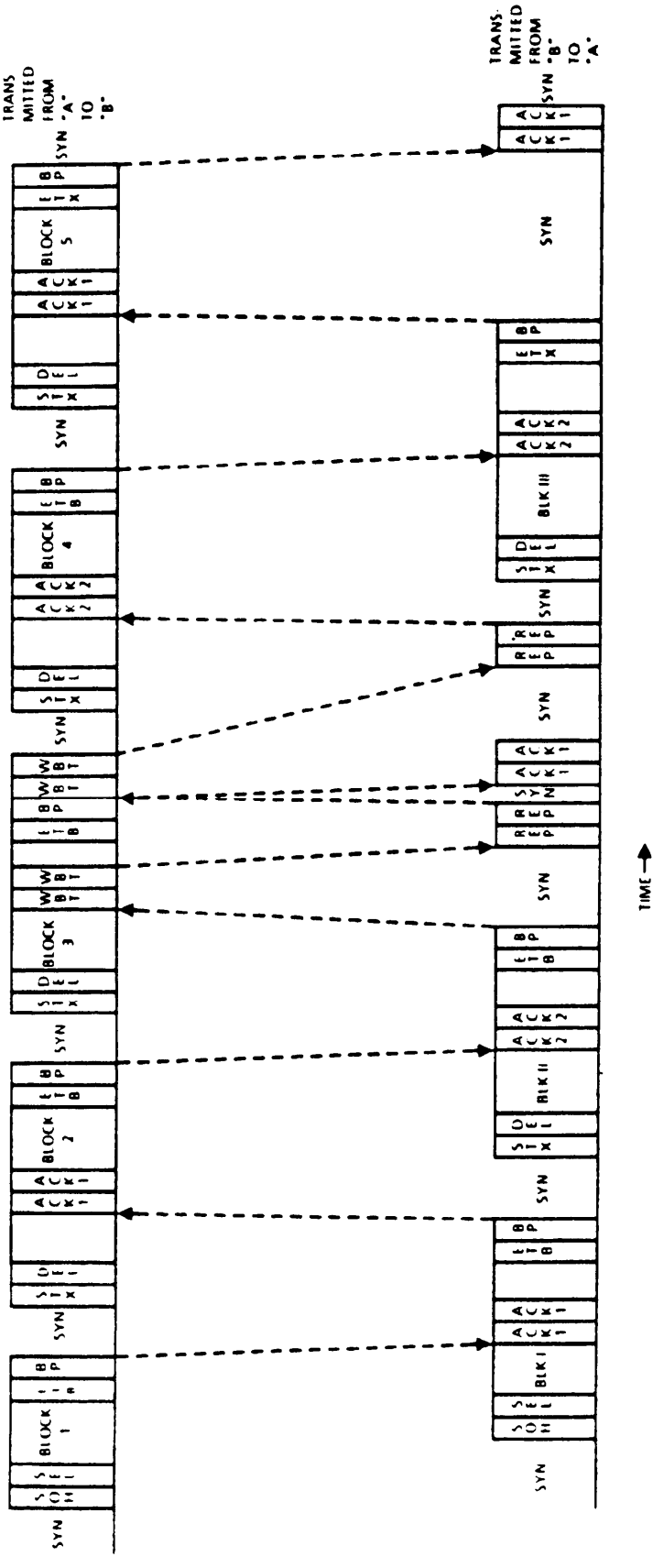


FIGURE 12. Mode I block-by-block operation - duplex - (error-free operation).

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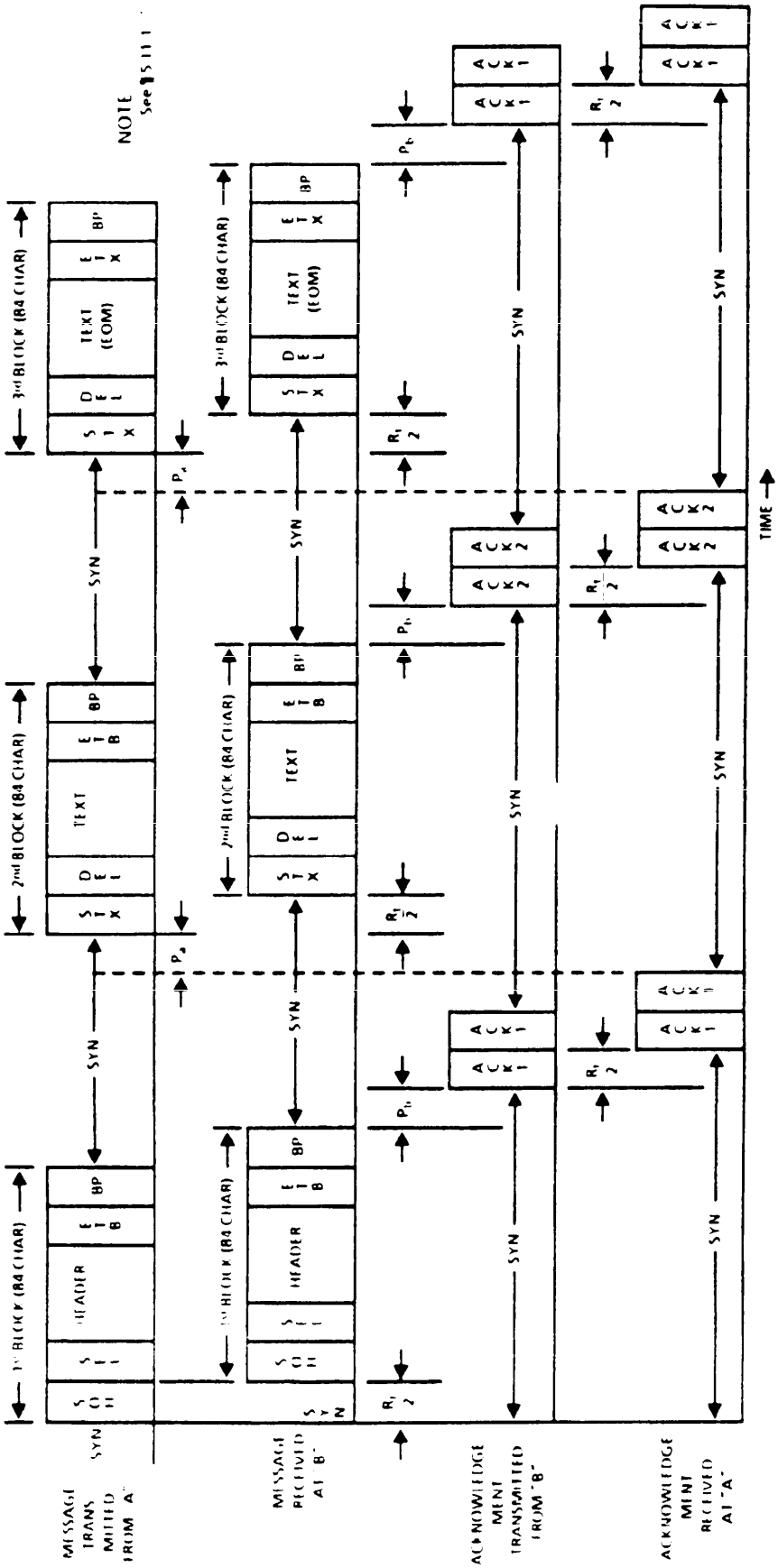


FIGURE 13. Mode I block-by-block operation-one direction (error-free operation).

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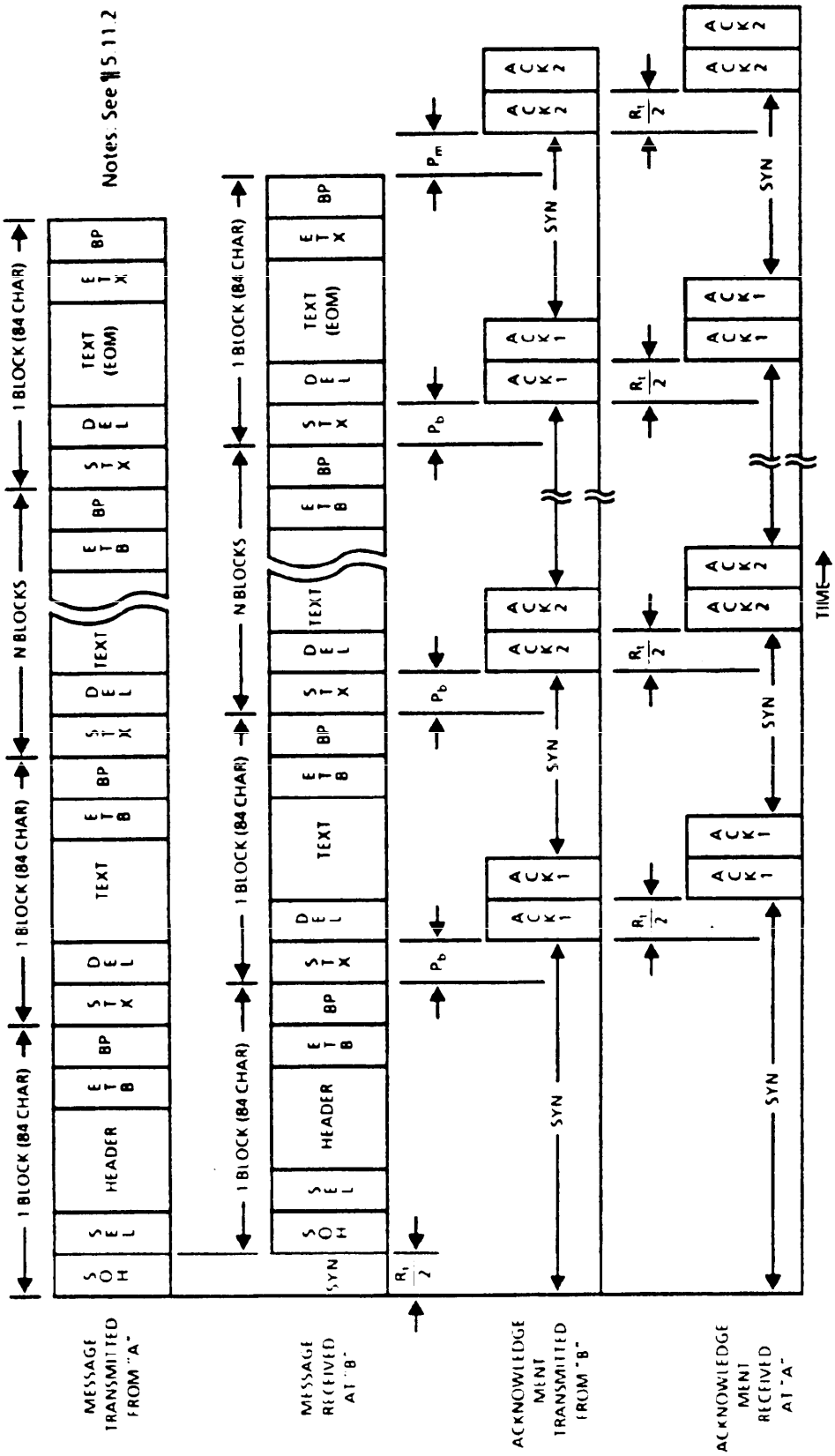


FIGURE 14. Mode I continuous operation one direction (error-free operation).

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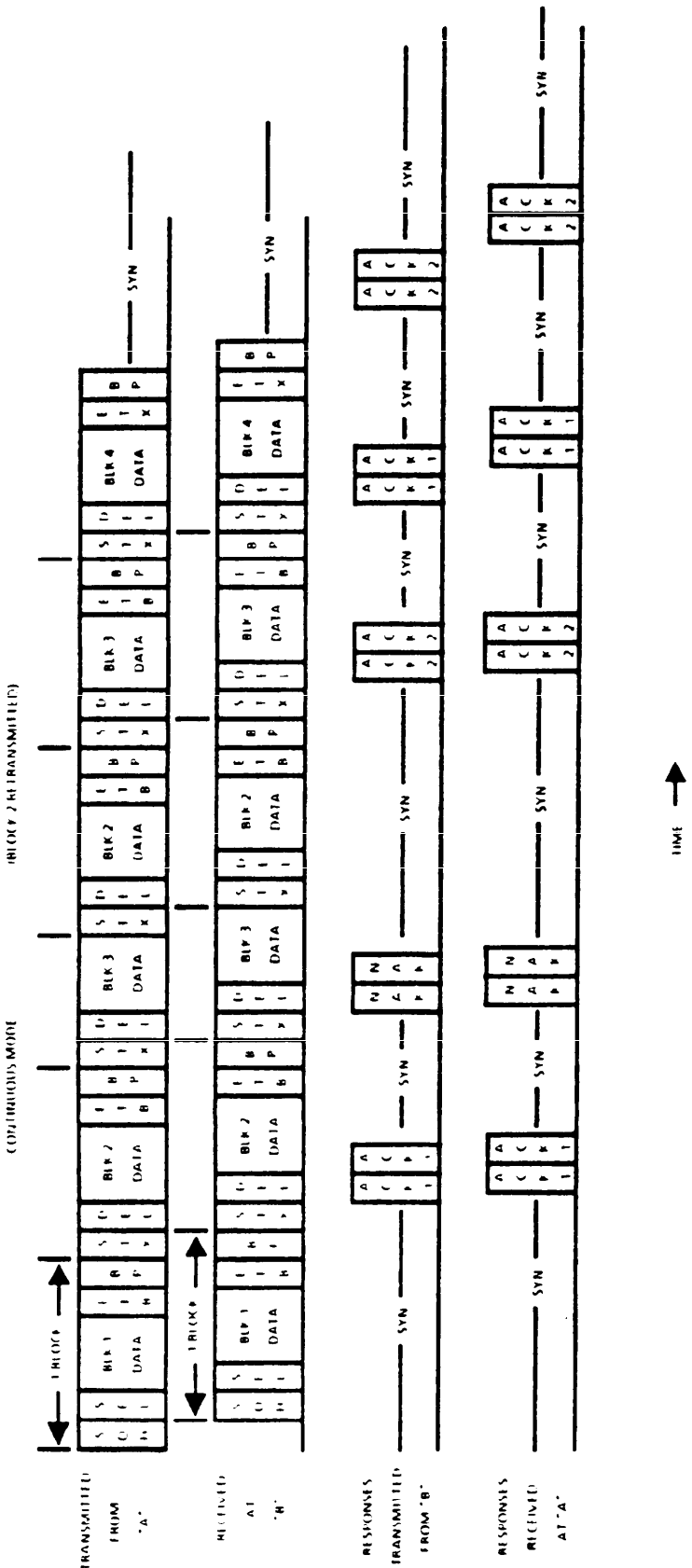
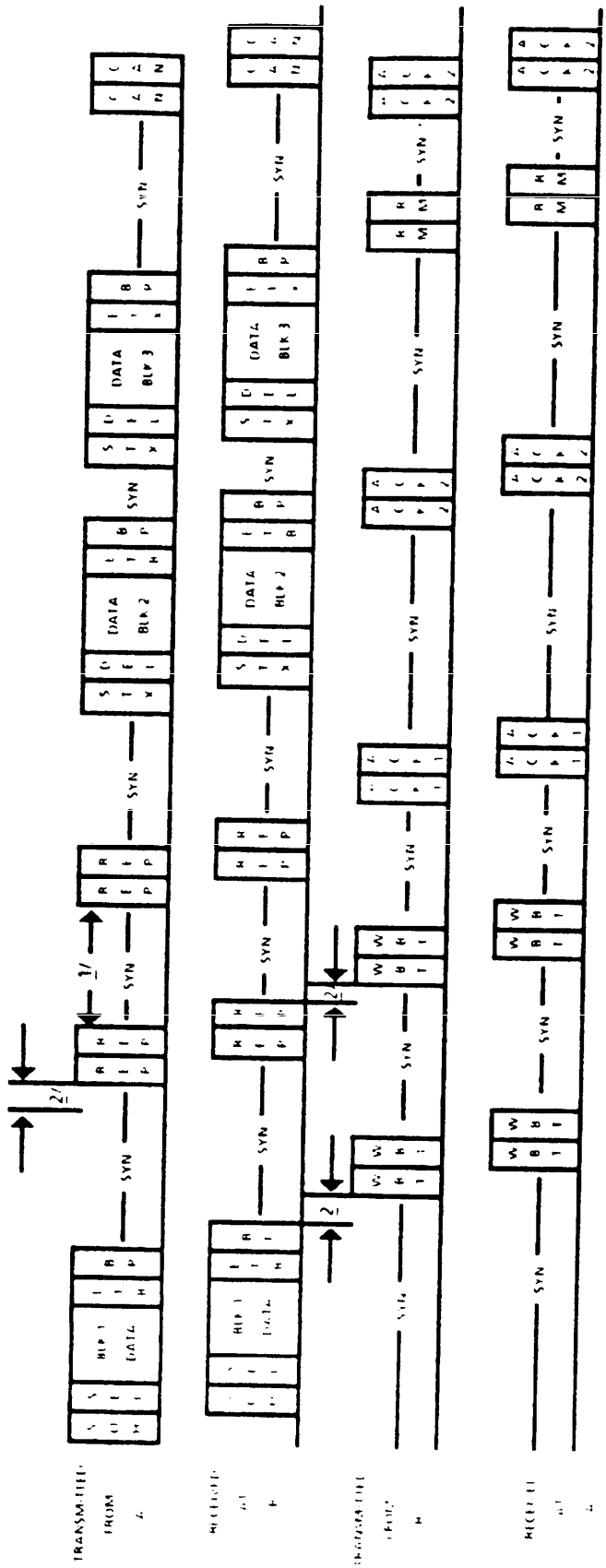


FIGURE 15. Mode I continuous mode - (block 2 received with errors).

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NOTE:
1/ = Transmitter answer timer
2/ = Reciever response time

FIGURE 16. Mode I block-by-block - REP-WBT-RM-CAN sequence.

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5.5 Framing characters. Framing characters shall delineate the beginning and end of each block. Two contiguous framing characters shall start the block, and two contiguous framing characters shall end the block (see figures 1 through 11). Framing characters shall have an even number of logical 1 bits except for block parity (BP). BP may be any one of the 256 octets (see 5.5.7).

5.5.1 Start of header (SOH). SOH shall be the first framing character of the first block of a message. There shall be only one SOH per message. The SOH shall be followed by a select (SEL) character. The sequence SOH/SEL shall not be split.

5.5.2 Select character (SEL). One of the select characters from table IV shall be the second framing character of the first block of a message. There shall be only one select character per message. The select character shall be used to indicate media and data formats.

5.5.3 Start of text (STX). STX shall be the first framing character of all blocks after the first block of a message. STX shall be followed by the delete (DEL) framing character. The sequence STX/DEL shall not be split.

5.5.4 Delete (DEL). DEL shall be placed in the second framing character position of all blocks, except for the first block of a message.

5.5.5 End of transmission block (ETB). ETB shall be the third framing character of all blocks except the last block of a message. ETB appears in the 83rd character position of a normal block or following end of medium (EM) in a short block. ETB shall be followed by BP. The sequence ETB/BP shall not be split.

5.5.6 End of text (ETX). ETX shall be the third framing character of the last block of a message. ETX shall be the 83rd character position of the last normal block of a message or following EM in the last short block of a message. ETX shall be followed by BP. The sequence ETX/BP shall not be split.

5.5.7 Block parity (BP). BP shall be the last framing character of every block in the message. BP shall follow ETB or ETX. BP shall be formed by the binary addition without carry of each bit position of a block starting with the second framing character. SOH and STX shall not be included in BP generation. BP may be any of the 256 octets of table II. When ETB or ETX is detected, the next character should be compared bit for bit with the receiver-generated BP character. If these two characters are not identical, the block shall be considered to have a parity error.

5.6 Control characters. Control characters are transmitted with an even number of logical 1 bits. There are three subsets of control characters:

a. Message,

b. Special, and

c. Synchronous idle.

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5.6.1 Message control characters. Message control characters shall control the exchange between the transmitting and receiving systems. There are two types of message control characters: receive control characters (RCC) and transmit control characters (TCC).

5.6.1.1 Receive control characters (RCC). RCC are sent by the receiver when the transmitter requests a response. All RCC are transmitted in identical contiguous pairs. When a request for an answer is received, the receiver shall respond with an answer before expiration of the response timer (see 5.9.1.2). RCC shall not be inserted between two framing characters, shall not be added to the block parity summation, and they shall not be counted in establishing the positional criteria for ETB/BP or ETX/BP. Requests for responses consist of:

a. Receipt of a properly framed block,

b. Receipt of the TCC cancel (CAN), and

c. Receipt of the TCC reply (REP).

5.6.1.1.1 Acknowledgment number one (ACK1). ACK1 shall be sent by the receiver to signal the transmitter that an error-free block has been received. ACK1 shall be one of the responses to a reply (REP). ACK1 shall be the answer to the first block received after a cancel (CAN) sequence. ACK1 shall be used alternately with ACK2 to indicate correctly received blocks. The transmitter shall recognize ACK1 when two contiguous ACK1 control characters are detected. The pattern of alternating ACK1 and ACK2 shall not be interrupted between messages; i.e., if the answer to the last block of a message was ACK1, then the answer to the first block of the following message shall be ACK2.

5.6.1.1.2 Acknowledgment number two (ACK2). ACK2 shall be sent by the receiver to signal the transmitter that an error-free block has been received. ACK2 shall be one of the responses to a REP. ACK2 shall be the answer to a CAN sequence. ACK2 shall be used alternately with ACK1 to indicate correctly received blocks. The transmitter shall recognize ACK2 when two contiguous ACK2 control characters are detected. The pattern of alternating ACK1 and ACK2 shall not be interrupted between messages; i.e., if the answer to the last block of a message was ACK1, then the answer to the first block of the following message shall be ACK2.

5.6.1.1.3 Negative acknowledgment (NAK). NAK shall be transmitted if an error has been detected in a properly framed block. NAK shall be one of the responses to a REP (see 5.6.1.2.1). It shall be sent when an answer is requested (after receipt of ETX/BP or ETB/BP), not at the time of error detection. Upon receipt of NAK, the transmitter shall retransmit the complete block for which the NAK applies. The transmitter shall recognize NAK when two contiguous NAK control characters are detected.

NOTE: An error is detected if any character in a data position of a properly framed block has an even number of logical 1 bits or if the calculated and received BP characters disagree.

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5.6.1.1.4 Reject message (RM). RM shall be transmitted as a response to a properly framed block to inform the transmitter that there is a defect in the message (such as an unacceptable routing indicator) that cannot be corrected by retransmission of the block, or that the receiver is unable to process the message. RM shall be one of the responses to a REP (see 5.6.1.2.1). Upon receipt of RM, the transmitter shall cause the message to be canceled. The transmitter shall recognize RM when two contiguous RM control characters are detected.

NOTE: Depending on the system and type of terminal, there may be limitations as to when an RM may be sent. A terminal is not permitted to send RM to an AUTODIN Switching Center, except after receipt of BP or REP following states such as power-up, power loss, and self-test.

5.6.1.1.5 Wait before transmitting (WBT). WBT shall be sent by the receiver in response to a properly framed block to indicate that it can no longer accept blocks. WBT shall be one of the responses to a REP (see 5.6.1.2.1). WBT shall also be an answer to a CAN sequence if the ACK2 cannot be sent before the expiration of the response timer (see 5.9.1.2). While WBT is being received, the transmitter shall send only synchronous idle (SYN) or other control characters. The transmitter shall recognize WBT when two contiguous WBT control characters are detected. If WBT is received by the transmitter for five minutes, then an alarm at the transmitter (of the message) shall be activated. The receiver clears WBT by the transmission of an ACK, NAK, or RM in response to a REP.

5.6.1.2 Transmit control characters (TCC). TCC shall be sent by the transmitter to direct the receiver to take some action. These characters are REP and CAN and shall be transmitted in identical contiguous pairs. Transmit control characters shall be transmitted only between blocks in block-by-block operation. In continuous operation, REP and CAN shall be sent between blocks, after the 80th data character position in a normal block, or immediately following EM in short blocks.

5.6.1.2.1 Reply (REP). REP shall be sent by the transmitter to direct the receiver to transmit its last response or its current updated response. As an example of a current updated response, the initial response could be WBT and the current updated response could be ACK1. The receiver shall recognize REP when two contiguous REP control characters are detected. The proper responses to REP are: ACK1, ACK2, NAK, RM, or WBT. If an incorrect ACK sequence is received in response to a REP sequence, the block shall be retransmitted. REP shall be used under the following conditions:.

a. REP shall be sent when the answer timer (see 5.9.1.1) expires if one of the following conditions exist:

   (1) ACK1, ACK2, NAK, or RM has not been received, or

   (2) WBT has been received.

b. During continuous mode operation and unless the response expected is for the final block of a message, the first REP shall not be sent until the 80th data character or EM of the following block has been sent.

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c. The answer timer shall be restarted each time REP is sent. When an ACK1, ACK2, NAK, or RM is received, the answer timer shall be stopped. If REP is sent three consecutive times without a response, then an alarm condition shall occur.

5.6.1.2.2 Cancel (CAN). CAN shall be sent by the transmitter to direct the receiver to cancel and discard the message currently being received or to synchronize the ACK sequence. The receiver shall recognize CAN when two contiguous CAN control characters are detected. If CAN is transmitted three consecutive times without a response, then an alarm condition shall occur. CAN shall be used under the following conditions:

a. The CAN sequence shall be transmitted whenever it becomes necessary to cancel transmission of the current message or to respond to RM.

b. CAN shall never be transmitted within the data portion of a block or when the answer to a block is outstanding. If the response to a block is WBT, CAN shall not be sent until a response of ACK1, ACK2, NAK, or RM has been received. CAN shall be acknowledged with ACK2. When CAN is sent, the answer timer shall be started. If ACK2 has not been received when the answer timer expires, CAN shall be sent again. If the response to CAN is WBT, CAN shall continue to be sent each time the answer timer expires.

c. If CAN is sent three times without an answer from the receiver, an alarm condition shall occur. The CAN sequence shall continue to be sent at the expiration of the answer timer.

d. A CAN shall be sent if an ACK1 or ACK2 is received out of sequence in direct response to a block (ETX/BP or ETB/BP).

5.6.2 Special control characters. Special control characters, which consist of EM, invalid sequence (INV), record mark (DC3), and mode change (MC), shall perform the following functions:

5.6.2.1 End of medium (EM). The EM control character shall mark the end of data in a block containing 79 or fewer data characters. EM shall be included in the block parity check and shall be followed by ETB/BP or ETX/BP.

5.6.2.2 Invalid sequence (INV). The INV shall be transmitted as two identical contiguous characters in response to each ACK1, ACK2, NAK, RM, or WBT sequence received when an answer is not expected by the transmitter. INV shall be recognized when two contiguous INV control characters are detected. An alarm condition shall occur when the INV is transmitted or received.

5.6.2.3 Mode change (MC). MC is a special control character used between terminal equipment to mark the beginning or the end of the binary or nonstandard code text portion of a message, or both. MC shall contain an even number of logical 1 bits. MC shall occur as the first character of a data block and shall be immediately followed by an EM (see 5.7.2 and figure 11).

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5.6.2.4 Record mark (DC3). DC3 is a special character defined for magnetic media terminals to enable transmission of single tape marks (or other single media separators) between magnetic media devices. DC3 characters contain an odd number of logical 1 bits and shall occur as the first character in a data block. DC3 shall be immediately followed by an EM (see 50.2.3 of appendix C):

5.6.3 Synchronous idle (SYN) control character. The SYN control character shall provide character synchronization between the transmitter and the receiver. SYN shall be transmitted continuously whenever the transmitter is not sending message blocks or control sequences.

5.7 Special short blocks. Mode I shall implement three special short blocks (see appendix C).

5.7.1 End of medium (EM) block. The EM block shall consist of five characters: STX, DEL, EM, ETB, and BP (see figure 10). This block is available for special applications, such as the end of header for magnetic media message transmission.

5.7.2 Mode change (MC) block. The MC block shall consist of six characters: STX, DEL, MC, EM, ETB, and BP (see figure 11). This block shall be used to define the beginning and end of the binary data portion of a message.

NOTE: Although MC is an even parity character, it shall be included in BP and shall not be considered a parity error.

5.8 Character synchronization. SYN shall be used to establish and maintain character synchronization when no other characters are being exchanged. Character synchronization shall be declared when four contiguous SYN characters are detected. The out-of-sync timer shall start when the block parity character is received. It shall be reset whenever a REP, CAN, properly framed block, or four consecutive SYN characters are received.

5.8.1 Excessive unexpected characters. The receiver shall consider the reception of a maximum of 170 contiguous unexpected characters as the system being out-of-sync. Once an out-of-sync condition is detected, the system shall alarm. The type of alarm and action to be taken shall be specified in the performance specification of the equipment.

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5.9 Mode I timing and time outs. Table V specifies the minimum Mode I requirements for timing and time out conditions and actions to be taken for proper Mode I operations.

5.9.1 Timers. There shall be a transmitter answer timer and a receiver response timer.

5.9.1.1 Transmitter answer timer. The answer timer shall be set to the maximum allowable time that the transmitter shall wait for an answer before requesting an answer. The precise time is a variable and includes: data rate, length of path, response time, etc. The answer timer shall never exceed five seconds. Nominally, the answer timer shall be in the range of 1.5 to 2.0 block transmission times. The transmitter answer timer shall be started when the BP of a block is transmitted, each time a CAN sequence is transmitted, and each time a REP sequence is transmitted. The timer shall be cleared only when an appropriate response is received (ACK1, ACK2, NAK, RM, or WBT).

5.9.1.2 Receiver response time. When a request for a response (ETX/BP) is received, the receiver shall respond within one second. When any other request for a response is received (ETB/BP, CAN, REP), the receiver shall respond within 3 character times (24 bits) for data rates less than 2400 bps. For data rates equal to or greater than 2400 bps, the receiver shall respond within 0.01 second.

5.9.2 Safe message store. A receiver shall positively acknowledge (ACK) the last block of a Mode I message (message acknowledgment) only after the message has been placed into safe message store,or a temporary storage element that is protected from loss of data, system perturbations for at least 24 hours, e.g., hard copy, paper tape, or magnetic media. The Mode I system shall place a message into safe message storage within one second after receipt of the ETX block.

5.10 Channel establishment/reestablishment. Channel establishment shall be performed in accordance with the procedures in tables VI and VII.

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TABLE V. Mode I timing and time outs.

FUNCTION		CAUSE	ACTION
RECEIVED	WBT	WBT received in response to a block	Send REP at answer timer interval.
		WBT received for 5 minutes	Alarm and send REP at answer timer interval.
		WBT received to REP	Send REP at answer timer interval.
		WBT received (no block/REP transmitted).	Alarm and send INV within 3 character (0.01 second) times.
	REP	REP received	Send ACK1, ACK2, NAK, WBT, or RM within 3 character times.
	RM	RM received	Alarm and send CAN within 3 character times.
	CAN	CAN received during message reception	Discard message and send ACK2 within 3 character times.
		CAN received between messages	Send ACK2 within 3 character times.
		CAN received; unable to send ACK2 within 3 character times.	Send WBT.
	NAK	NAK received to transmitted block	Resend NAKed block within 3 character times.
		NAK received (no block transmitted)	Alarm and send INV within 3 character times.
	INV	INV sequence received	Alarm.
	ETB/BP	Block received, no errors	Send appropriate ACK within 3 character times.
		Block received - error detected	Send NAK within 3 character times.
		Block received - unable to accept more data.	Send WBT.
		Block received - not in ACK SYNC (initial start-up).	Send RM.

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TABLE V. Mode I timing and timeouts - Continued.

FUNCTION		CAUSE	ACTION
RECEIVED	ETX/BP	Block received No errors - message validated (if appropriate)	Place message into safe storage - send appropriate ACK within 1 second.
		Block received - error in block	Send NAK within 3 character times.
		Block received - no errors in block - message validation fails	Send RM.
		Block received - unable to reply within 1 second	Send WBT.
		Block received - not in ACK syn (Initial Start-up)	Send RM.
	ACK 1	ETX/BP transmitted	Proper ACK received - send next message if appropriate. Wrong ACK received - send CAN within 3 character times.
		ETB/BP transmitted	Proper ACK received - send next block. Wrong ACK received - send CAN within 3 character times.
		CAN transmitted	Send CAN at answer timer interval until ACK 2 received.
		REP transmitted	ACK 1 expected - send next block.
			ACK 2 expected - retransmit last block.
		WBT transmitted	Send WBT at answer timer intervals or appropriate response if able to accept data.
		RM transmitted	Until CAN is received - resend RM at answer timer interval.

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TABLE V. Mode I timing and time outs - Continued.

FUNCTION		CAUSE	ACTION
RECEIVED	ACK 2	ETX/BP transmitted	ACK 2 expected - send next message. ACK 1 expected - send CAN.
		ETB/BP transmitted	ACK 2 expected - send next block. ACK 1 expected - send CAN.
		CAN transmitted	Resend message.
		REP transmitted	ACK 2 expected - send next block. ACK 1 expected - retransmit last block.
		WBT transmitted	Send WBT at answer timer interval or appropriate response if able to accept data.
		RM transmitted	Retransmit RM at answer timer intervals.
TRANSMITTED	REP	REP transmitted	Accept only SYN, WBT, ACK1, ACK2, RM, or NAK.
		WBT received	Send REP at answer timer interval
		ACK received (expected)	Clear REP - send next block or SYN
		ACK received (not expected)	Retransmit last block. Clear the REP answer timer
		RM received (no message being sent)	Alarm - send CAN at answer timer interval until ACK2 received
		RM received (intramessage)	Alarm - send CAN at answer timer interval until ACK2 received, then resend message
		NAK received (intramessage)	Retransmit last block - clear REP answer timer.
		CAN received	Send ACK2 - send REP at answer timer interval.
		INV received	Alarm - send REP at answer timer interval.

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TABLE V. Mode I timing and time outs - Continued.

FUNCTION		CAUSE	ACTION
TRANSMITTED	WBT	WBT transmitted	Accept only SYN, REP, CAN, or RCC.
		Block received Unable to send ACK or NAK	Send WBT and ignore block.
		CAN received Unable to send ACK2	Send WBT.
		REP received Able to accept data	Send appropriate response.
		REP received Unable to accept any more data	Send WBT.
	RM	RM transmitted	Accept only SYN, REP, CAN, or RCC.
		CAN received	Alarm - discard message and send ACK2.
		REP received	Send RM.
		INV received	Alarm.
	CAN	CAN transmitted	Accept SYN, ACK2, WBT, or data - alarm.
		CAN transmitted 3 times without response	Alarm - send CAN at answer timer interval.
		ACK2 received	Resend message if appropriate - clear answer timer.
		WBT received	Send CAN at answer timer interval.
		WBT received for 5 minutes	Alarm and send CAN at answer timer interval.
		No response	Send CAN at answer timer interval.
		ACK1 received	Send INV and Alarm and send CAN at answer or timer interval.
		REP received	Alarm and send CAN at answer timer interval.
		INV received	Alarm - send CAN at answer timer interval.

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TABLE V. Mode I timing and time outs - Continued.

FUNCTION		CAUSE	ACTION
TRANSMITTED	ETB/BP	Block transmitted	Accept SYN, ACK1, ACK2, NAK, WBT, or TCC.
	(ETX/BP)	ACK received (expected)	Send next block (next message).
		ACK received (not expected)	Alarm and send CAN.
		NAK received	Retransmit block.
		RM received	Alarm and send CAN.
		No response	Send REP at answer timer interval.
		WBT received	Send REP at answer timer interval.

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TABLE VI. Mode I operations.

STATE	Receiver (RCVR)	Transmitter (XMTR)	Description
TRAFFIC: IDLE	SYN →	← SYN	No data to send. XMTR and RCVR send SYN New circuit established by XMTR.
CHANNEL ESTABLISHMENT	ACK2 →	← CAN	XMTR sends CAN sequence.
			RCVR detects CAN and sends ACK2 RCVR clears any incomplete messages.
		← SYN	XMTR received ACK2.
	SYN →		RCVR and XMTR are in character and ACK synchronization (sync).
DATA TRANSFER		← Transmit SOH block	XMTR begins transfer of first message block.
	ACK1 →		RCVR receives first block without error and sends ACK1.
		← Transmit second block	XMTR receives ACK1 and sends next block.
	NAK →		RCVR detects error in block and sends NAK.
		← Retransmit second block	XMTR receives NAK and retransmits 2nd block of message.
	ACK2 →		RCVR receives 2nd block of message without error and sends ACK2
		← Transmit ETX block	XMTR receives ACK2 and sends next block which is ETX block
	ACK1 →		RCVR receives ETX block without error and sends ACK1
IDLE	SYN →	← SYN	XMTR and RCVR send SYN to maintain synchronization (sync)

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TABLE VII. Mode I - reject and cancel message procedures.

	STATE	Receiver (RCVR)	Transmitter (XMTR)	Description
TRAFFIC	IDLE	SYN →	← SYN	No data to send. New circuit established by XMTR
	CHANNEL ESTABLISHMENT REJECT MESSAGE		← Transmit MSG block	XMTR begins transfer of MSG blocks before ACK sync established
		RM →		Since ACK sync has not been established, RCVR rejects incoming MSG and forces XMTR to send CAN sequence.
			← CAN	XMTR detects RM and cancels MSG. RCVR detects CAN, discards MSG being received, and sends ACK2.
		ACK2 →		
		SYN →	← SYN	XMTR and RCVR are now in character and ACK sync.
	MESSAGE CANCELLATION		← Transmit MSG block	XMTR begins transfer of MSG blocks
		SYN/NAK/ACK →		RCVR sends response to each received MSG block as blocks are received
			← Transmit ETX block	
		SYN/NAK/ACK →		
			← CAN	XMTR detects uncorrectable error in MSG or response received and cancels MSG.
		ACK2 →		RCVR detects CAN, discards MSG being received, and sends ACK2
		SYN →	← SYN	XMTR and RCVR sends SYN to maintain synchronization
	REJECT MESSAGE		← Transmit MSG block	XMTR begins transfer of MSG blocks
		SYN/NAK/ACK →		RCVR sends response to each MSG block as blocks are received.
			← Transmit ETX block	XMTR sends last block of MSG
		RM →		RCVR detects uncorrectable error in MSG and sends RM sequence
			← CAN	XMTR detects RM and cancels MSG
		ACK2 →		RCVR detects CAN, discards MSG being received, and sends ACK2
	IDLE	SYN →	← SYN	XMTR and RCVR send SYN to maintain synchronization

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5.11 Mode I throughput calculations. The following equations shall be used to calculate the theoretical throughput time to transmit Mode I messages. It is assumed that no errors are generated or detected.

5.11.1 Block-by-block operation. The following equation shall be used to calculate the theoretical time to transmit a block-by-block Mode I message, where:

t = Time, in seconds, to transmit a Mode I message

B = Number of bits necessary to transmit a Mode I message (including framing and EM characters)

R = Data rate in bits per second (bps)

N = Number of blocks necessary to transmit the message

P_a = Time, in seconds, required to process ACK, equal to or less than 0.01 second for R equal to or greater than 2400 bps and $\frac{24}{R}$ for R less than 2400 bps

P_b = Time, in seconds, required to process block, equal to or less than 0.01 second for R equal to or greater than 2400 bps and $\frac{24}{R}$ for R less than 2400 bps

[NOTE: 24 = Three character times (24 bits) to respond to a command (ETB/BP, etc.)]

16 = (2 characters ACK seq) x (8 bits/character) = 16 bits

R_t = Round-trip delay in seconds

R_T = $R_t + P_a + P_b$

P_m = Time, in seconds, required to process message, equal to or less than 1.0 second

then:

$$t = \left[\frac{B}{R} + N\left(P_a + P_b + \frac{16}{R}\right) + N(R_t) + P_m\right]$$

$$t = \left[\frac{B}{R} + N\left(\frac{16}{R}\right) + N(P_a + P_b + R_t) + P_m\right]$$

$$t = \left[\frac{B + N(16)}{R} + N(R_T) + P_m\right] \text{ seconds}$$

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5.11.2 Continuous operation. The following equation shall be used to calculate the theoretical time to transmit a continuous Mode I message, where:

t = Time, in seconds, to transmit a Mode I message

B = Number of bits necessary to transmit a Mode I message (including framing and EM characters)

R = Data rate in bits per second (bps)

R_t = Round-trip delay in seconds

16 = (2 characters ACK seq) x 8 bits/character = 16 bits

P_a = Time, in seconds, required to process ACK, equal to or less than 0.01 second for R equal to or greater than 2400 bps and $\frac{24}{R}$ for R less than 2400 bps

P_b = Time, in seconds, required to process block, equal to or less than 0.01 second for R equal to or greater than 2400 bps and $\frac{24}{R}$ for R less than 2400 bps

[NOTE: 24 = Three character times (24 bits) to respond to a command (ETB/BP, ETX/BP, etc.)]

P_m = Time required to process message, equal to or less than 1.0 second

n = Total number of blocks in the message

B_i = Number of bits in the ith block (including framing and EM characters)

R_T = $R_t + P_a + P_b$

then:

$$t = \left[\left(\frac{B}{R} + R_t + \frac{16}{R} + P_a + P_b + P_m\right) + \sum_{i=2}^{i=n} \left(\frac{\left|R_t + \frac{16}{R} + P_a + P_b - \left(\frac{B_i - 16}{R}\right)\right| + \left(R_t + \frac{16}{R} + P_a + P_b - \left(\frac{B_i - 16}{R}\right)\right)}{2}\right)\right]$$

$$t = \left[\left(\frac{B + 16}{R} + R_T + P_m\right) + \sum_{i=2}^{i=n} \left(\frac{\left|R_T - \left(\frac{B_i - 32}{R}\right)\right| + \left(R_T - \left(\frac{B_i - 32}{R}\right)\right)}{2}\right)\right] \text{ seconds}$$

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5.11.3 Examples. The following subparagraphs are examples of the time to transmit a message containing 6900 data characters over an error-free duplex channel for block-by-block and continuous Mode I operation for R_t = 0 second, 0.5 second, and 1.0 second. Table VIII lists the time to transmit block-by-block and continuous Mode I messages for the three R_t's and various data rates (R).

5.11.3.1 Transmission time, block-by-block operation. The following example illustrates the calculation of the theoretical time to transmit a message containing 6900 data characters using Mode I block-by-block channel control procedures, where:

B = 57,992 bits

N = 87 blocks

R = 2400 bits per second

R_T = 0.02 second (R_t = 0)

P_m = 1 second

then:

$$t = \left[\frac{B + N(16)}{R} + N(R_T) + P_m \right]$$

$$t = \left[\frac{57992 + 87(16)}{2400} + 87(0.02) + 1.0 \right]$$

$$t = \left[\frac{57992 + 1392}{2400} + 1.74 + 1 \right]$$

$$t = \left[24.75 + 2.74 \right]$$

t = 27.49 seconds

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5.11.3.2 Transmission time, continuous operation. The following example illustrates the calculation of the theoretical time to transmit a message containing 6900 data characters using Mode I continuous channel control procedures, where:

B = 57992 bits

R = 2400 bits

R_t = 0.02 second (R_t = 0)

P_m = 1 second

n = 87 blocks

B_i for i = 2 through 86 = 672

B_i (for i = 87) = 200 bits

P_a = P_b = 0.01 second

then:

$$t = \left[\left(\frac{B+16}{R} + R_T + P_m\right) + \sum_{i=2}^{i=n} \left(\frac{\left|R_T - \left(\frac{B_i - 32}{R}\right)\right| + \left(R_T - \left(\frac{B_i - 32}{R}\right)\right)}{2}\right)\right]$$

$$t = \left[\left(\frac{57992+16}{2400} + 0.02 + 1\right) + 85 \left(\frac{\left|0.02 - \frac{672-32}{2400}\right| + \left(0.02 - \frac{672-32}{2400}\right)}{2}\right) + \left(\frac{\left|0.02 - \frac{200-32}{2400}\right| + \left(0.02 - \frac{200-32}{2400}\right)}{2}\right)\right]$$

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$$t = \left[\left(\frac{58008}{2400} + 1.02\right) + 85\left(\frac{\left|0.02 - \frac{640}{2400}\right| + \left(0.02 - \frac{640}{2400}\right)}{2}\right) + \left(\frac{\left|0.02 - \frac{168}{2400}\right| + \left(0.02 - \frac{168}{2400}\right)}{2}\right)\right]$$

$$t = \left[\left(24.17 + 1.02\right) + 85\left(\frac{|0.02 - .267| + (0.02 - .267)}{2}\right) + \frac{|0.02 - 0.07| + (0.02 - 0.07)}{2}\right]$$

$$t = \left[25.19 + 85\left(\frac{0.265 - 0.265}{2}\right) + \left(\frac{0.05 - 0.05}{2}\right)\right]$$

$$t = \left[25.19 + 85(0) + 0\right]$$

$$t = 25.19 \text{ seconds}$$

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TABLE VIII. Mode I transmission time in seconds.

Mode I Transmission Time in Seconds (t) for a 6900-character message						
Data Rate (bps) (R)	R_t = 1 second		R_t = 0.5 second		R_t = 0 second	
	B-B 1/	Cont 2/	B-B	Cont	B-B	Cont
16000	93.46	89.96	49.96	46.46	6.46	4.66
9600	95.93	90.10	52.43	46.60	8.93	7.07
8000	97.17	90.17	53.67	46.67	10.17	8.28
4800	102.12	90.46	58.62	46.96	15.12	13.11
4000	104.59	90.60	61.09	47.10	17.59	15.53
2400	114.49	91.18	70.99	47.68	27.49	25.19
2000	119.78	91.81	76.28	48.31	32.78	30.03
1200	140.97	94.35	97.47	50.85	53.97	49.38
600	193.94	100.70	150.44	98.56	106.94	97.76
300	299.87	196.12	256.37	195.12	212.87	194.52
150	511.74	389.24	468.24	388.54	424.74	388.04
75	935.47	776.08	891.97	775.58	848.47	775.08

NOTES:

1/ B-B = Block-by-block operation

2/ Cont = Continuous operation

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Key word listing. The following key words and phrases apply to MIL-STD-188-116-1:

AUTODIN
Synchronous
Channel coordination procedure
Data communication protocol
Eight-bit code
Mode I
Protocol
Block-by-block
Continuous

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APPENDIX A

DEPARTMENT OF DEFENSE DIRECTIVE 4640.11
21 DECEMBER 1987
SUBJECT: MANDATORY USE OF MILITARY TELECOMMUNICATIONS STANDARDS IN THE MIL-STD-188 SERIES

This appendix contains information related to MIL-STD-188-116-1. Appendix A is a mandatory part of this standard.

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APPENDIX A

Department of Defense
DIRECTIVE

December 21, 1987
NUMBER 4640.11

USD(A)

SUBJECT: Mandatory Use of Military Telecommunications Standards in the MIL-STD-188 Series

References: (a) DoD Directive 5137.1, "Assistant Secretary of Defense (Command, Control, Communications, and Intelligence)," April 2, 1985
(b) DoD Directive 4120.3, "Defense Standardization and Specification Program," February 10, 1979
(c) DoD 4120.3-M, "Defense Standardization and Specification Program Policies, Procedures and Instructions," August 1978, authorized by DoD Directive 4120.3, February 10, 1979
(d) through (l), see enclosure 1

A. PURPOSE

This Directive establishes policy governing the application and use of the MIL-STD-188-100, -200, and -300 series of telecommunications standards; prescribes procedures; and assigns responsibilities.

B. APPLICABILITY AND SCOPE

1. This Directive applies to the Office of the Secretary of Defense (OSD), the Military Departments, the Organization of the Joint Chiefs of Staff (OJCS), the Unified and Specified Commands, and the Defense Agencies (hereafter referred to collectively as "DoD Components").

2. Its provisions cover the development and application of military telecommunications standards, specifically, the following:

a. MIL-STD-188-100 series, containing standards common to long-haul and tactical communications.

b. MIL-STD-188-200 series, containing standards exclusive to tactical telecommunications.

c. MIL-STD-188-300 series, containing standards exclusive to long-haul telecommunications.

C. POLICY

It is DoD policy that the interoperability and performance standards in the MIL-STD-188 series that are required for interoperability and compatibility of DoD telecommunications equipment and systems are mandatory for use for all inter- and intra-DoD Component systems and equipment.

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APPENDIX A

D. RESPONSIBILITIES

1. The Assistant Secretary of Defense (Command, Control, Communications, and Intelligence) (ASD(C^{3}I)) shall interact with affected DoD Components under DoD Directive 5137.1 (reference (a)).

2. The Assistant Secretary of Defense for Production and Logistics (ASD(P&L)) is responsible for Defense Standardization and Specification Program (DSSP) policy, administration, and guidance.

3. The Director, Standardization and Data Management (SDM), Office of the Deputy Assistant Secretary of Defense (Production Support) (ODASD(PS)), manages and administers the DSSP and establishes policy, program guidance, and controls under DoD Directive 4120.3 (reference (b)).

4. The Director, Defense Standardization Program Office (DSPO), Defense Product Standards Office (DPSO), Defense Data Management Office (DDMO), ODASD(PS), shall assist the Director, SDM, in managing and administering the DoD communications standardization program for developing and establishing DSSP policies, program guidance, and controls.

5. The Heads of DoD Components shall:

a. Comply with this Directive, so that:

(1) Developers of the MIL-STD-188 series ensure that each standard is not only essential but of uniformly high quality, clear and concise as to application suitable for use in acquisition packages and, to the maximum extent possible, compatible with existing or proposed national and international (both Government and non-Government) telecommunications standards.

(2) Users of these standards cite in their procurement specifications only those standards essential to the proper functioning of the device or system over its projected lifetime.

b. Ensure the application of the MIL-STD-188 series in their organic acquisition specifications.

c. Be responsible for policing and enforcing the use of the MIL-STD-188 series standards within the DoD Component.

d. Support the development, revision, and use of the MIL-STD-188 series documents and, when necessary, provide personnel and funding resources.

e. Incorporate in each activity's internal review process a method for ensuring that the telecommunications standards are referenced to the extent necessary in acquisition documents.

f. Be the granting authority for waivers and deviations for intra-DoD Component systems and equipment, and shall forward any consideration of and granting of waivers and deviations to the standardization office responsible for the maintenance of the MIL-STD-188 series standard concerned.

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g. Ensure that their standardization offices forward waivers and deviations for intra-DoD Component systems and equipment to the Defense Communications Agency (DCA) and the Joint Tactical Command, Control, and Communications Agency (JTC3A).

6. Director, Defense Communications Agency, and Director, Joint Tactical Command, Control, and Communication Agency, as the lead activities for the MIL-STD-188 series of standards, shall be the granting authorities for waivers and deviations for inter-DoD Component systems and equipment and shall review intra-DoD waivers and deviations granted by DoD Components.

E. PROCEDURES

1. DSSP. Under DoD Directive 4120.3 and DoD 4120.3-M (references (b) and (c)), the objectives of the DSSP are to improve the operational readiness of the DoD Components and ensure the cost-effectiveness of systems and equipment. The SDM is responsible for administering and managing the DSSP, which provides a uniform series of specifications, standards, and related documents. Under the DSSP, DoD Components develop military specifications and standards related to equipment acquisitions, including telecommunications equipment.

2. Tailoring of Military Standards. It is neither cost- nor mission-effective to impose military standards on all systems and equipment specifications blindly without consideration and review of each individual case. The applicability and relevance of the standard to the intended use of the equipment must be considered. DoD Directive 5000.43 (reference (d)) outlines DoD policies for the selection and tailoring of specifications used for acquisition. To prevent the misapplication of standards, the Department of Defense classifies most military standards as "approved for use," rather than "mandatory for use."

3. Application of Standards in System Acquisition. The required telecommunications standards selected from the MIL-STD-188 series shall be included, as appropriate, in defense acquisition documents (e.g., Statements of Work (SOWs), etc.). This direction is in addition to the authority and procedures contained in references (b), (c), (d) and DoD Directive 5000.1 and DoD Instruction 5000.2 (references (e) and (f)). DoD Directive 4630.5 (reference (g)), DoD Directive 5105.19 (reference (h)), and DoD Directive 5154.28 (reference (i)) highlight the development and application of interoperability and compatibility standards in the areas of tactical long-haul telecommunications.

4. Interoperability and Standardization

a. As stated in DoD Directives 2010.6, 2010.7, and 3100.4 (references (j) through (l)), the United States shall maximize the utility and effectiveness of allied telecommunications resources through equipment interoperability and standardization.

b. The MIL-STD-188 series addresses telecommunications design parameters, influences the functional integrity of telecommunications systems and their ability to interoperate efficiently with other functionally similar Government and commercial systems, and shall be mandatory for use within the Department of Defense.

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APPENDIX A

F. EFFECTIVE DATE AND IMPLEMENTATION

This Directive is effective immediately. Forward one copy of implementing documents to the Assistant Secretary of Defense (Command, Control, Communications, and Intelligence) within 120 days.

William H. Taft, IV
Deputy Secretary of Defense

Enclosure - 1
References

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REFERENCES, continued

(d) DoD Directive 5000.43, "Acquisition Streamlining," January 15, 1986
(e) DoD Directive 5000.1, "Major and Non-Major Defense Acquisition Programs," September 1, 1987
(f) DoD Instruction 5000.2, "Defense Acquisition Program Procedures," September 1, 1987
(g) DoD Directive 4630.5, "Compatibility and Interoperability of Tactical Command and Control, Communications, and Intelligence Systems," October 9, 1985
(h) DoD Directive 5105.19, "Defense Communications Agency (DCA)," August 10, 1978
(i) DoD Directive 5154.28, "Joint Tactical Command, Control, and Communications Agency (JTC^3A)," July 5, 1984
(j) DoD Directive 2010.6, "Standardization and Interoperability of Weapons Systems and Equipment within the North Atlantic Treaty Organization," March 5, 1980
(k) DoD Directive 2010.7, "Policy on Rationalization of NATO and NATO Member Telecommunications Facilities," July 6, 1981
(l) DoD Directive 3100.4, "Harmonization of Qualitative Requirements for Defense Equipment of the United States and Its Allies," September 27, 1963

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APPENDIX B

ABBREVIATIONS AND ACRONYMS

This appendix provides definitions of abbreviations and acronyms used in this document. Appendix B is a nonmandatory part of this document.

ACK	Positive acknowledgment
ADCCP	Advanced Data Communication Control Procedures
ARQ	Automatic Repeat-Request
ASCII	American Standard Code for Information Interchange
B	Total number of bits, including SOH/STX, ETX/ETB, SEL, DEL, EM, and BP, to transmit a Mode I message.
Bi	Total number of bits in ith block.
BP	Block parity
bps	bits per second
CAN	Cancel
CHAR	Character
DC3	Device Control 3
DCA	Defense Communications Agency
DCAC	Defense Communications Agency Circular
DCS	Defense Communications System
DEL	Delete
DoD	Department of Defense
DoDISS	DoD Index of Specifications and Standards
EM	End of medium
EOT	End of transmission
ETB	End of transmission block
ETX	End of text
INV	Invalid sequence
JCS	Joint Chiefs of Staff
JTCO	Joint Tactical Communications Office
LSB	Least significant bit
MC	Mode change
MSB	Most significant bit
MSG	Message

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NAK	Negative acknowledgment
NMCS	National Military Command System
OUSD	Office of the Under Secretary of Defense
R	Data rate in bps
RCC	Receive control characters
RCVR	Receiver
REP	Reply
RM	Reject message
s	Second(s)
SEL	Select character
SOH	Start of header
STX	Start of text
SYN	Synchronous idle
SYNC	Synchronization
TCC	Transmit control characters
TRI-TAC	Tri-Service Tactical Communications (JTCO)
WBT	Wait before transmit
XMTR	Transmitter

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APPENDIX C

DATA CONVERSION TECHNIQUES

This appendix contains information in support of MIL-STD-188-171 and is mandatory to the extent specified in the procurement document.

10. GENERAL

10.1 Scope. This appendix defines techniques that may be used to transmit data, using Mode I, by systems using codes other than odd parity ASCII or the Mode I character set.

10.2 Application. The methodology presented in section 50 shall be used in magnetic tape conversion and shall be used as a guide in developing information interchange techniques among systems that do not use Mode I and ASCII codes.

20. REFERENCE DOCUMENTS. Not applicable.

30. DEFINITIONS. Not applicable.

40. GENERAL REQUIREMENTS

40.1 Minimum criteria. Any system or terminal transmitting information using the Mode I channel coordination procedure shall conform to the following minimum criteria:

a. The smallest data element shall be the octet.

b. The octets within the data portion of each block shall contain an odd number of logical 1 bits.

c. Framing characters and control characters shall be the character set defined in this standard.

d. The special block defined as mode change (MC) shall be used to defi the beginning and end of other data exchange codes and binary bit streams (see 5.3.3.3).

40.2 User responsibility. When using codes other than odd parity ASCII or the Mode I character sets in switched systems such as AUTODIN, the user has the following responsibilities to:

a. Conform to the basic data formats of the system.

b. Ensure that the intended recipient has the capability to decipher, decode, or otherwise reconstruct the nonstandard coded data.

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50. SPECIFIC METHODOLOGY

50.1 Introduction. The guidance in this section describes a methodology for seven- and nine-track magnetic tape conversions that have been successfully employed in AUTODIN. This methodology may be used to develop new methodologies (see figures 17-28).

50.2 Magnetic tape messages. This section contains the interface and control criteria for variable-length data records and nonstandard code messages as applied to seven- and nine-track magnetic tape operation. It applies to terminals receiving and transmitting structured and nonstructured magnetic tape formatted messages. A magnetic tape message is a series of contiguous characters recorded on and read from magnetic tape as a single unit. Messages are separated by interrecord gaps. The length of messages is variable. Whenever a magnetic tape message is greater than 80 characters, the message is formatted in 80-character blocks for transmission by the transmitter and reformatted by the receiver. The end of a magnetic tape message is indicated by EM. The EM is transmitted after the last character of the message. If the last block of the message contains 80 data characters, the EM is sent in a block by itself following the last block of the message (see figures 10, 22, and 26).

50.2.1 Structured format. Structured format magnetic tape data may contain one or more messages. The message header and end of transmission (EOT) on structured format magnetic tape are contained on the magnetic tape. A double tape mark is written on the tape after the last message to indicate the end of data to be transmitted.

50.2.2 Nonstructured format. The nonstructured format magnetic tape data contains only one message, and all information is transmitted as data. The message header and EOT on nonstructured format data are introduced from another source. Additional information may be introduced from a separate source. After header transmission, the text is read from magnetic tape. The message text on magnetic tape is terminated by a double tape mark. The nonstructured format allows data to be transmitted from tapes that are void of communication headers or EOTs, and it allows data to be written on the receiving tape in the same manner.

50.2.3 Tape marks

50.2.3.1 Single tape mark. A tape mark is used as a "data separator" for separating label records from data records, separating individual files, or separating groups of records within a particular file. Single tape marks shall be transmitted by formatting the DC3 character in a special short block of six characters (see figure 28).*

* The DC3 block shall consist of six characters: STX, DEL, DC3, EM, ETB, and BP (see figure 28). This block shall be transmitted whenever a single tape mark (or other single media separator) is read within a message text.

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50.2.3.2 Double tape mark. Double tape marks are used as follows.

a. On tapes containing one or more structured format messages, a double tape mark indicates that no more messages are contained on the tape. Double tape mark sequences are not permitted as text separators.

b. In nonstructured format, the double tape mark shall indicate the end of message text.

c. The double tape mark sequence (see a and b above) shall not be transmitted.

50.2.4 Code translation

50.2.4.1 Seven-track tape (SEL character B). A seven-track tape message consists of seven bits per track: six bits of binary data plus one parity bit. The terminal, in preparing the seven-track magnetic tape data for transmission, converts the magnetic tape seven-bit pattern to the eight-bit Mode I data characters as shown in table IX.

50.2.4.2 Nine-track tape (SEL character C). A nine-track tape message consists of nine-bits per track: eight bits of binary data plus one parity bit. The terminal converts the nine-track tape text data into Mode I block data characters as follows (see figure 17):

a. Converts the nine bits of text of the magnetic tape data into eight bits by removing the parity bit,

b. Segments the magnetic tape into groups of three text characters (24 bits),

c. Converts the 24-bit groups into four six-bit groups, and

d. Converts the six bits of each group into the Mode I data character in the same manner as the six data bits of the seven-track data are converted.

NOTE: If there are not three eight-bit characters on the last character conversion for a record, logical 1 bits shall be inserted to fill the last six-bit character before adding the EM character.

50.2.5 Transmission sequences. Figures 18 through 27 illustrate the manner in which different magnetic tape messages (records) are converted into blocks and transmitted.

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50.2.5.1 Header. Format lines that constitute the header shall use data characters. Select Character B is used for seven-track tape and Select Character C is used for nine-track tape. If the last block of the record containing the header contains fewer than 80 characters, an EM control character is transmitted after the last character of the record (see figure 6). The EM is followed by the framing characters ETB/BP. If the last block of the header record contains 80 characters, then the EM control character is transmitted in a special short block (see figure 10).

50.2.5.2 Text. The beginning of binary text is indicated by transmission of the control character mode change (MC) in a special short block (see figure 11). The magnetic tape text blocks are taken from tape and converted for transmission into 80-character line blocks. The character DC3 is transmitted in a special block (see figure 28) and is formed whenever a single tape mark record is read within the message text of either a structured or nonstructured tape. The end of binary text is indicated by the transmission of the MC block. If the last block of a magnetic tape record contains fewer than 80 characters, an EM control character is transmitted after the last character of the record (see figure 8). The EM is followed by ETB/BP. If the last block of the magnetic tape record contains 80 characters, the EM control character is transmitted in a special short block.

50.2.5.3 End of transmission (EOT). The last block to be transmitted is the EOT block that contains the ETX framing character.

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TABLE IX. Conversion from seven-track tape to Mode I data (block) character.

Seven-Track Magnetic Tape								Mode I Block Character							
BITS								BITS							
b_7	b_6	b_5	b_4	b_3	b_2	b_1		a_8	a_7	a_6	a_5	a_4	a_3	a_2	a_1
P	0	0	0	0	0	0		0	1	0	0	0	0	0	0
P	0	0	0	0	0	1		1	1	0	0	0	0	0	1
P	0	0	0	0	1	0		1	1	0	0	0	0	1	0
P	0	0	0	0	1	1		0	1	0	0	0	0	1	1
P	0	0	0	1	0	0		1	1	0	0	0	1	0	0
P	0	0	0	1	0	1		0	1	0	0	0	1	0	1
P	0	0	0	1	1	0		0	1	0	0	0	1	1	0
P	0	0	0	1	1	1		1	1	0	0	0	1	1	1
P	0	0	1	0	0	0		1	1	0	0	1	0	0	0
P	0	0	1	0	0	1		0	1	0	0	1	0	0	1
P	0	0	1	0	1	0		0	1	0	0	1	0	1	0
P	0	0	1	0	1	1		1	1	0	0	1	0	1	1
P	0	0	1	1	0	0		0	1	0	0	1	1	0	0
P	0	0	1	1	0	1		1	1	0	0	1	1	0	1
P	0	0	1	1	1	0		1	1	0	0	1	1	1	0
P	0	0	1	1	1	1		0	1	0	0	1	1	1	1
P	0	1	0	0	0	0		1	1	0	1	0	0	0	0
P	0	1	0	0	0	1		0	1	0	1	0	0	0	1
P	0	1	0	0	1	0		0	1	0	1	0	0	1	0
P	0	1	0	0	1	1		1	1	0	1	0	0	1	1
P	0	1	0	1	0	0		1	1	0	1	0	1	0	0
P	0	1	0	1	0	1		0	1	0	1	0	1	0	1
P	0	1	0	1	1	0		1	1	0	1	0	1	1	0
P	0	1	0	1	1	1		0	1	0	1	0	1	1	1
P	0	1	1	0	0	0		0	1	0	1	1	0	0	0
P	0	1	1	0	0	1		1	1	0	1	1	0	0	1
P	0	1	1	0	1	0		1	1	0	1	1	0	1	0
P	0	1	1	0	1	1		0	1	0	1	1	0	1	1
P	0	1	1	1	0	0		1	1	0	1	1	1	0	0
P	0	1	1	1	0	1		0	1	0	1	1	1	0	1
P	0	1	1	1	1	0		0	1	0	1	1	1	1	0
P	0	1	1	1	1	1		1	1	0	1	1	1	1	1

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TABLE IX. Conversion from seven-track tape to Mode I data (block) character - Continued.

Seven-Track Magnetic Tape								Mode I Block Character							
BITS								BITS							
b_7	b_6	b_5	b_4	b_3	b_2	b_1		a_8	a_7	a_6	a_5	a_4	a_3	a_2	a_1
P	1	0	0	0	0	0		1	1	1	0	0	0	0	0
P	1	0	0	0	0	1		0	1	1	0	0	0	0	1
P	1	0	0	0	1	0		0	1	1	0	0	0	1	0
P	1	0	0	0	1	1		1	1	1	0	0	0	1	1
P	1	0	0	1	0	0		0	1	1	0	0	1	0	0
P	1	0	0	1	0	1		1	1	1	0	0	1	0	1
P	1	0	0	1	1	0		1	1	1	0	0	1	1	0
P	1	0	0	1	1	1		0	1	1	0	0	1	1	1
P	1	0	1	0	0	0		0	1	1	0	1	0	0	0
P	1	0	1	0	0	1		1	1	1	0	1	0	0	1
P	1	0	1	0	1	0		1	1	1	0	1	0	1	0
P	1	0	1	0	1	1		0	1	1	0	1	0	1	1
P	1	0	1	1	0	0		1	1	1	0	1	1	0	0
P	1	0	1	1	0	1		0	1	1	0	1	1	0	1
P	1	0	1	1	1	0		0	1	1	0	1	1	1	0
P	1	0	1	1	1	1		1	1	1	0	1	1	1	1
P	1	1	0	0	0	0		0	1	1	1	0	0	0	0
P	1	1	0	0	0	1		1	1	1	1	0	0	0	1
P	1	1	0	0	1	0		1	1	1	1	0	0	1	0
P	1	1	0	0	1	1		0	1	1	1	0	0	1	1
P	1	1	0	1	0	0		1	1	1	1	0	1	0	0
P	1	1	0	1	0	1		0	1	1	1	0	1	0	1
P	1	1	0	1	1	0		0	1	1	1	0	1	1	0
P	1	1	0	1	1	1		1	1	1	1	0	1	1	1
P	1	1	1	0	0	0		1	1	1	1	1	0	0	0
P	1	1	1	0	0	1		0	1	1	1	1	0	0	1
P	1	1	1	0	1	0		0	1	1	1	1	0	1	0
P	1	1	1	0	1	1		1	1	1	1	1	0	1	1
P	1	1	1	1	0	0		0	1	1	1	1	1	0	0
P	1	1	1	1	0	1		1	1	1	1	1	1	0	1
P	1	1	1	1	1	0		1	1	1	1	1	1	1	0
P	1	1	1	1	1	1		0	1	1	1	1	1	1	1

NOTE: b_7 of seven-track tape is the magnetic tape character parity bit a_7 of the Mode I character and is always a 1.

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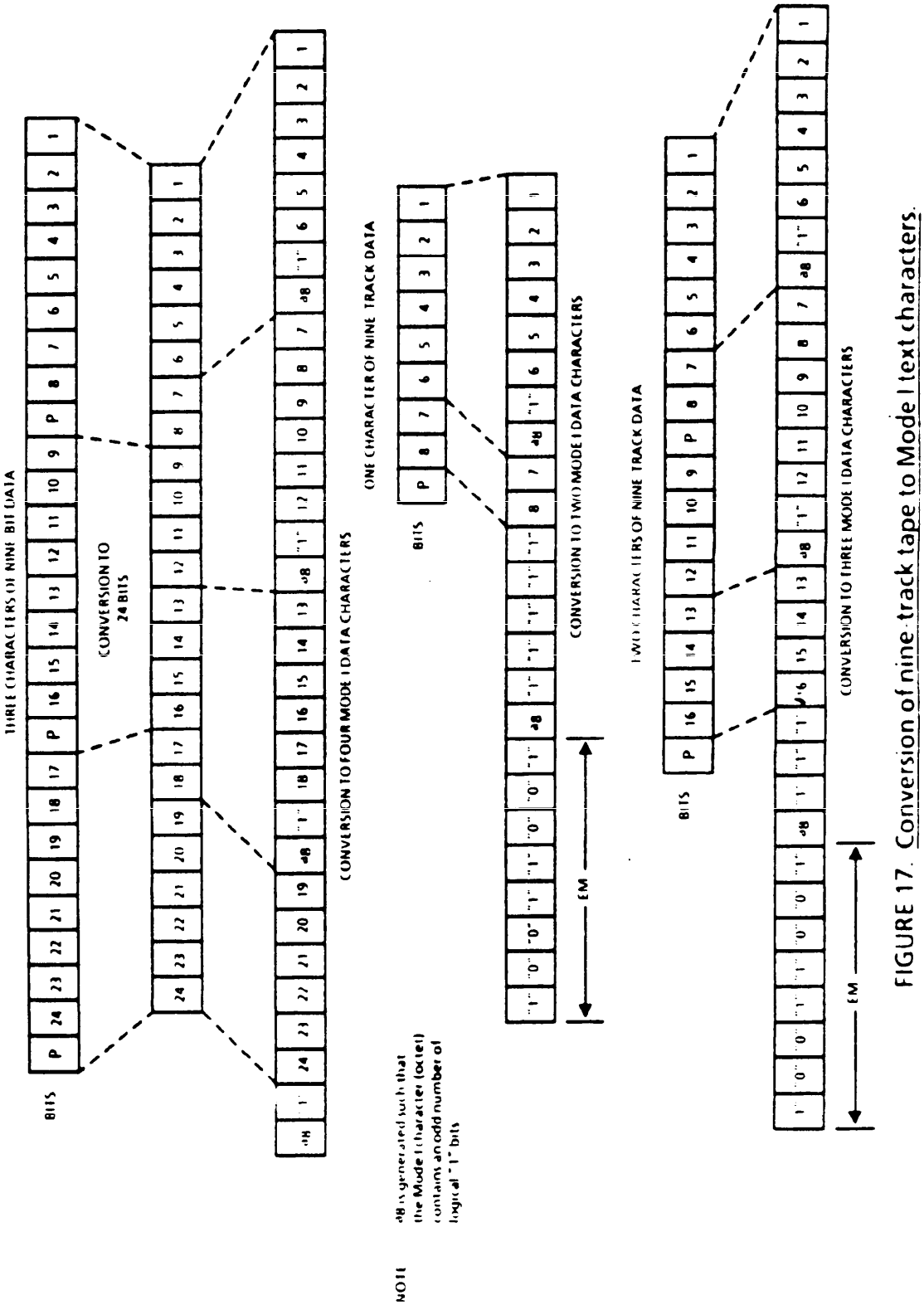
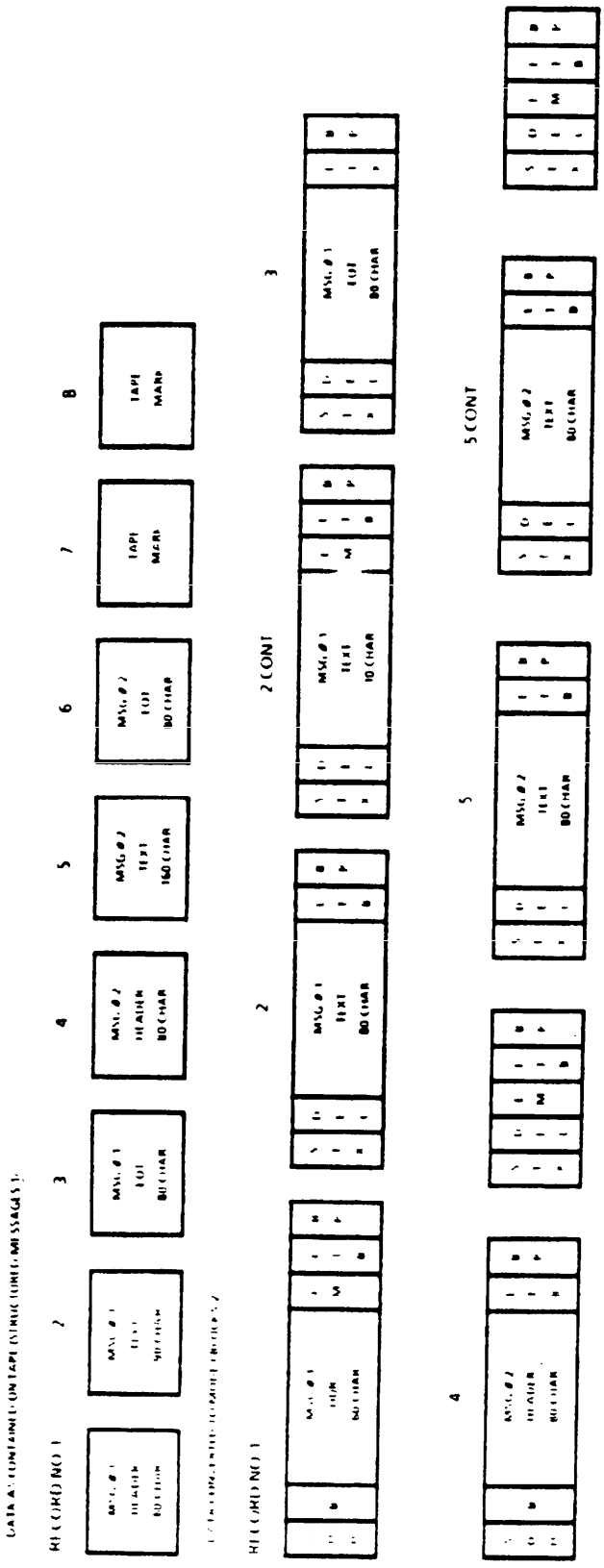


FIGURE 17. Conversion of nine-track tape to Mode I text characters.

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NOTE: 1/ MAGNETIC TAPE CONTAINING ONE OR MORE STRUCTURED MESSAGES IN LOCAL OR NATIVE CHARACTER CODING SCHEME WITH VARIABLE-LENGTH RECORDS. HEADER, TEXT, AND EOT RECORDS FOR EACH MESSAGE ON TAPE.

2/ SEE FIGURE 19 FOR TRANSMISSION OF THESE MODE I BLOCKS.

FIGURE 18. Typical Mode I structured format for seven-track tape without a single-tape mark.

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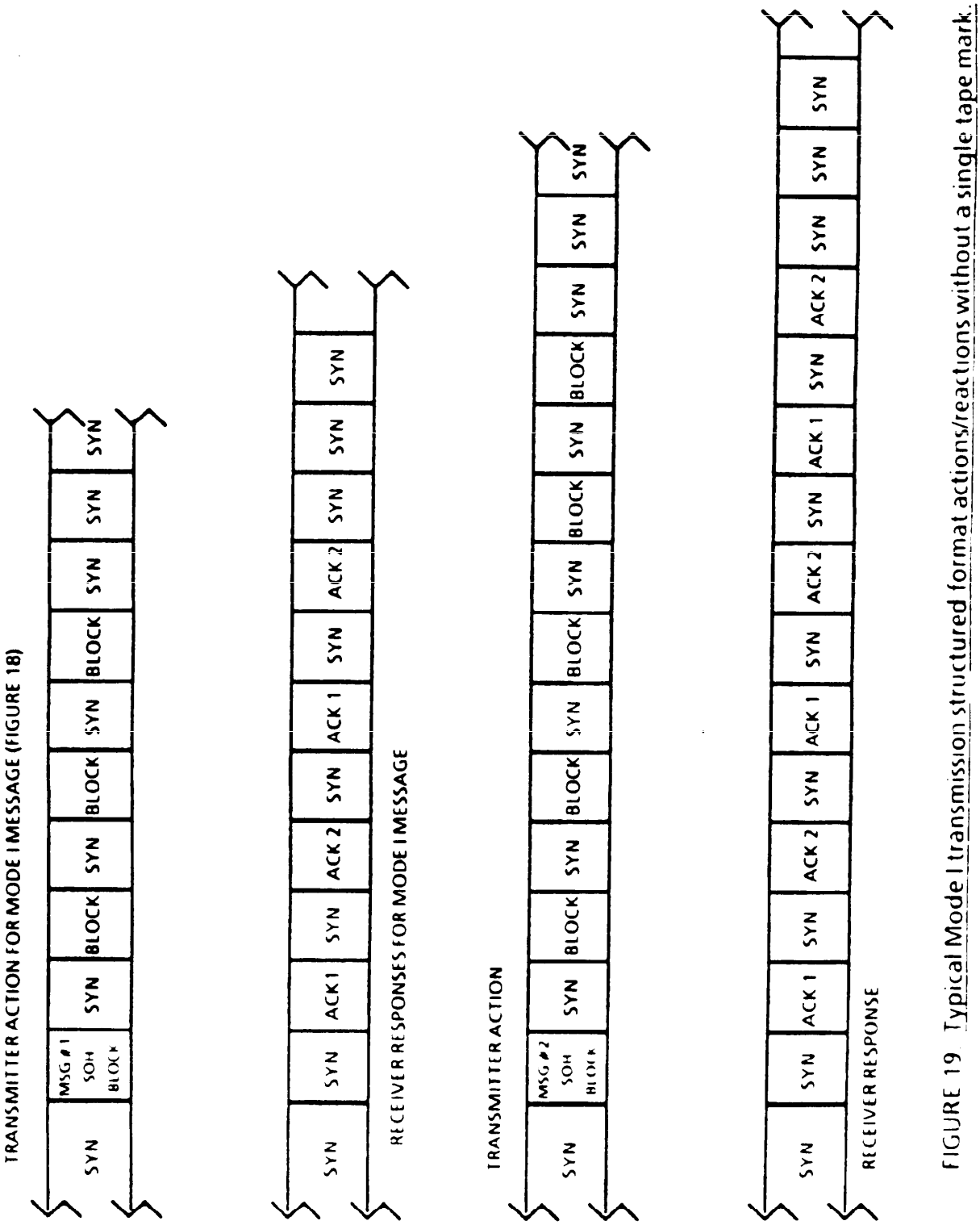
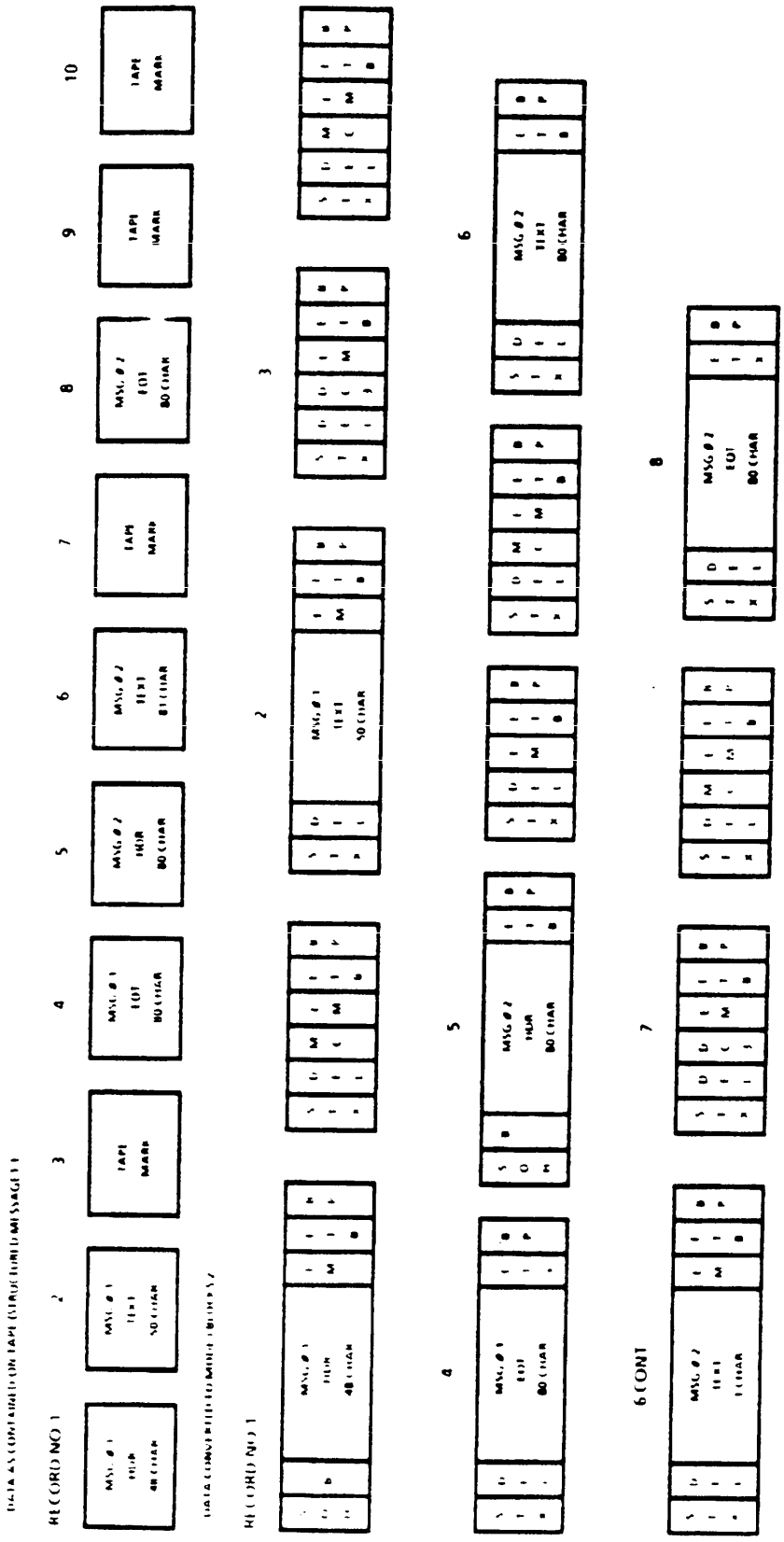


FIGURE 19 Typical Mode I transmission structured format actions/reactions without a single tape mark.

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NOTES: 1/ MAGNETIC TAPE CONTAINING ONE OR MORE STRUCTURED BINARY STREAM MESSAGES WITH VARIABLE-LENGTH RECORDS HEADER, TEXT, AND EOT RECORDS FOR EACH MESSAGE ON TAPE.

2/ SEE FIGURE 21 FOR TRANSMISSION OF THESE MODE I BLOCKS.

FIGURE 20. Typical Mode I structured format for seven-track tape with a tape mark.

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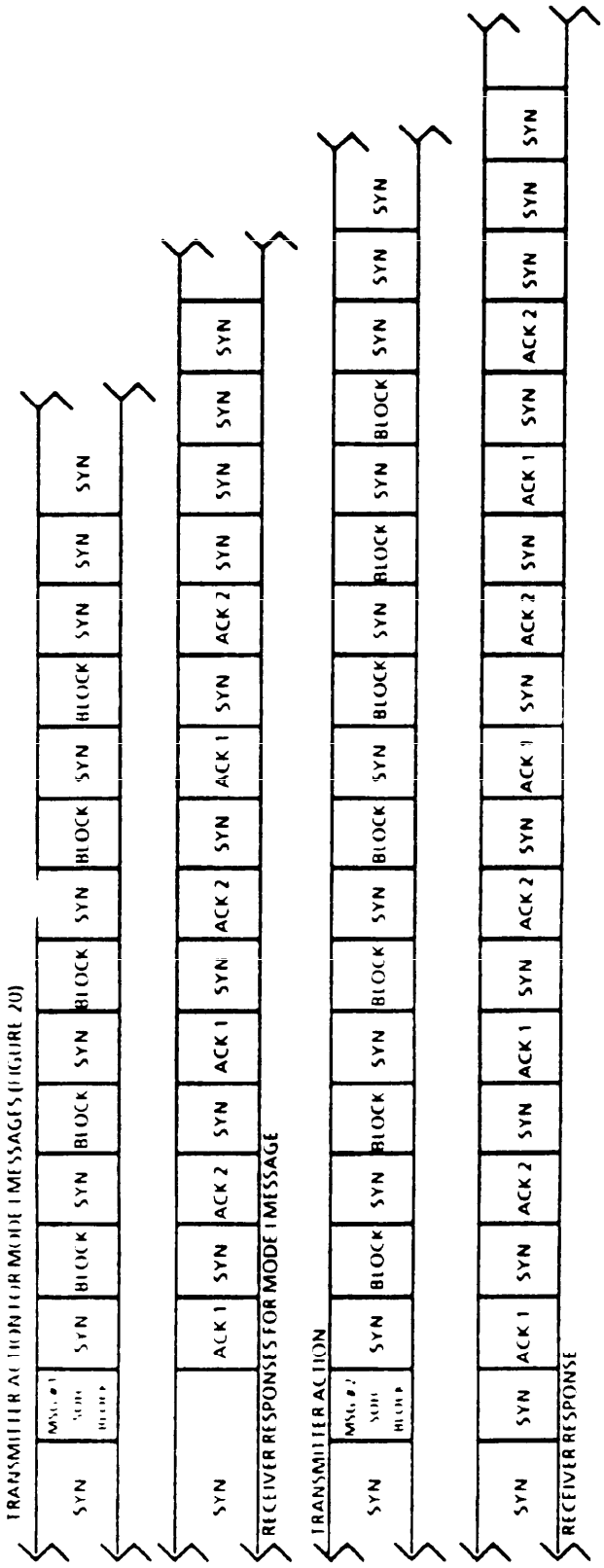
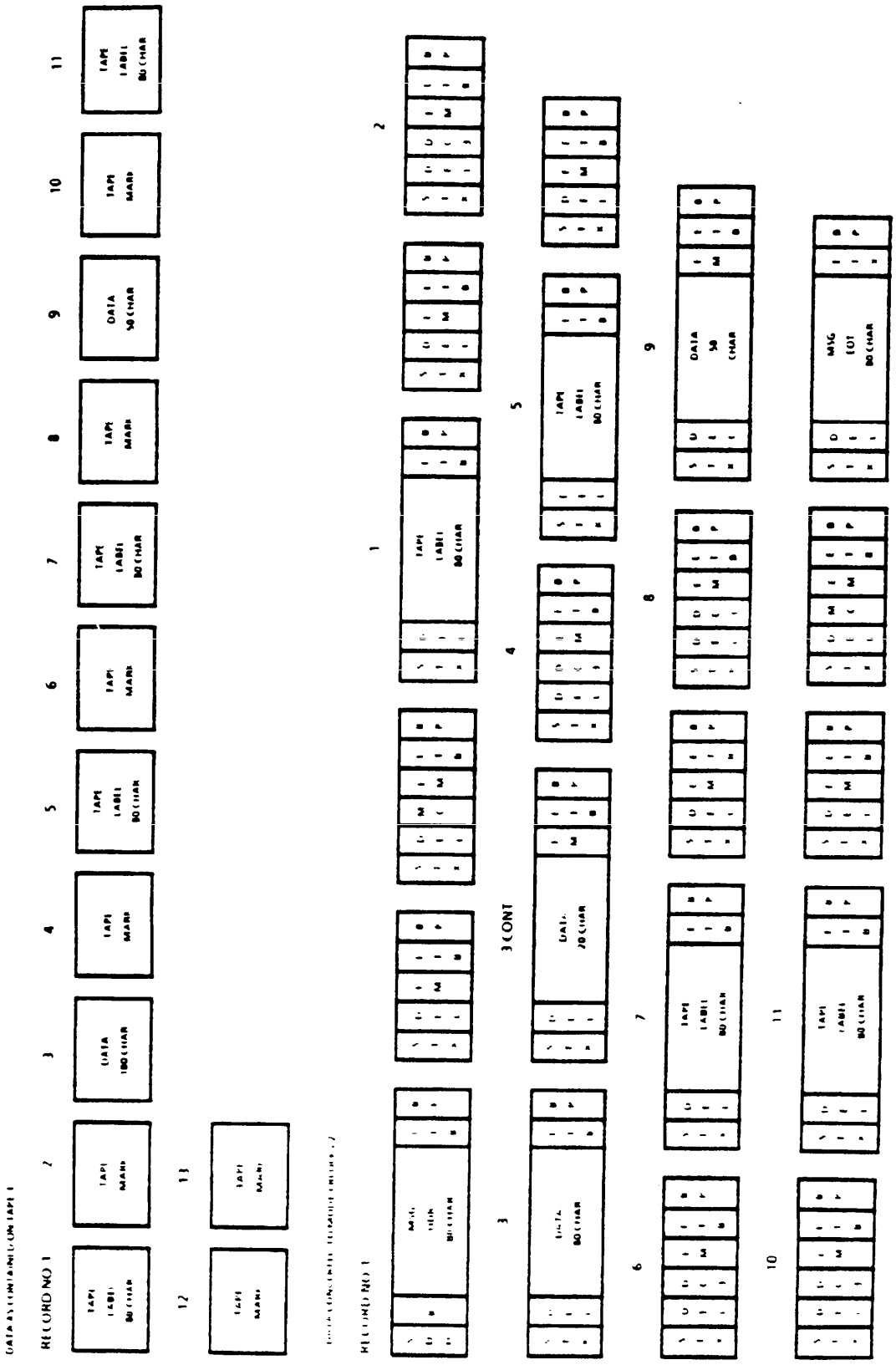


FIGURE 21. Mode I transmission structured format action/reaction with a tape mark.

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NOTES 1/ MAGNETIC TAPE CONTAINING ONLY ONE NONSTRUCTURED BINARY STREAM MESSAGE WITH VARIABLE LENGTH RECORDS. HEADER AND EOT RECORDS NOT ON TAPE

2/ SEE FIGURE 23 FOR TRANSMISSION OF THESE MODE I BLOCKS

FIGURE 22. Mode I nonstructured format for seven-track tape - mode change.

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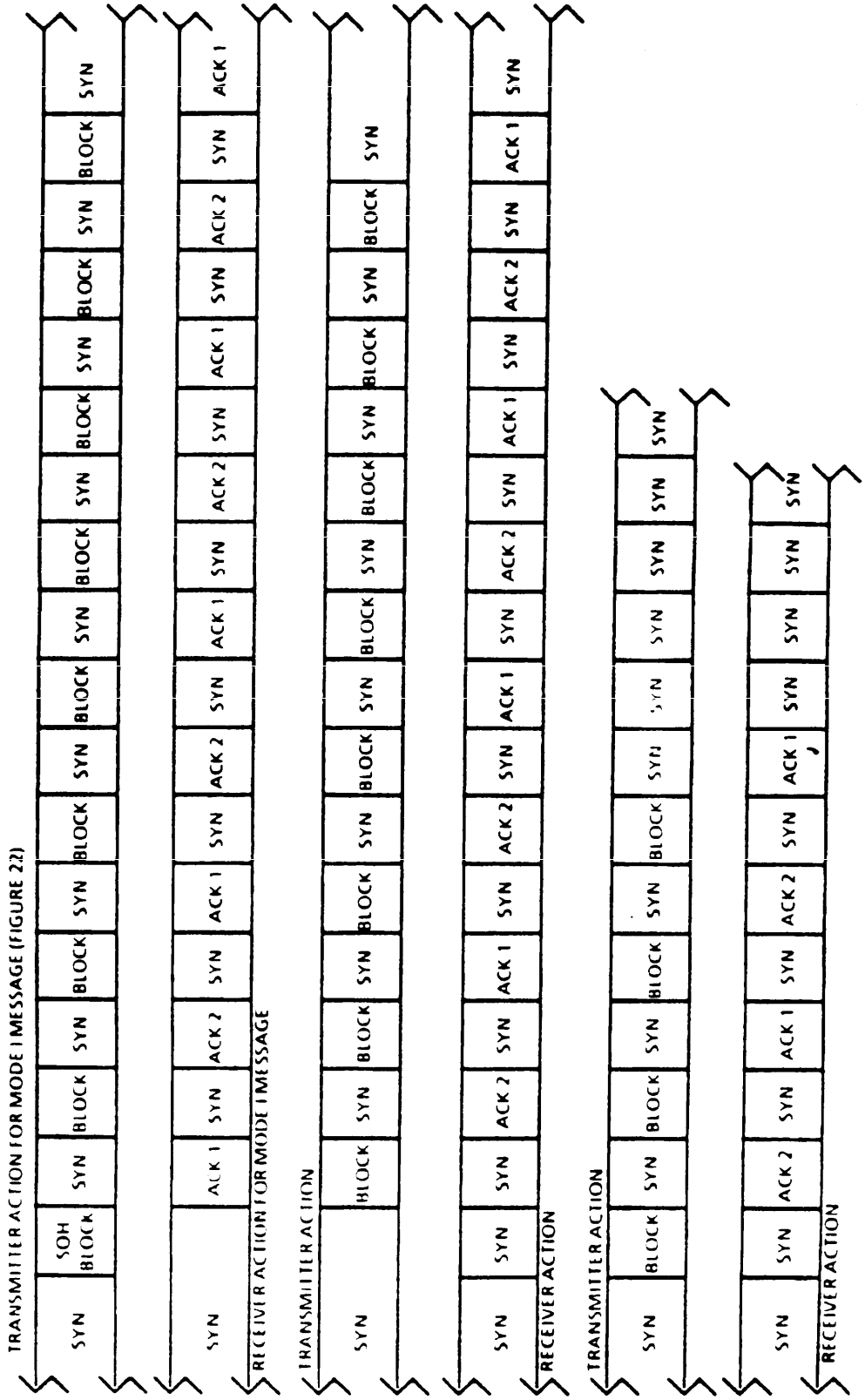
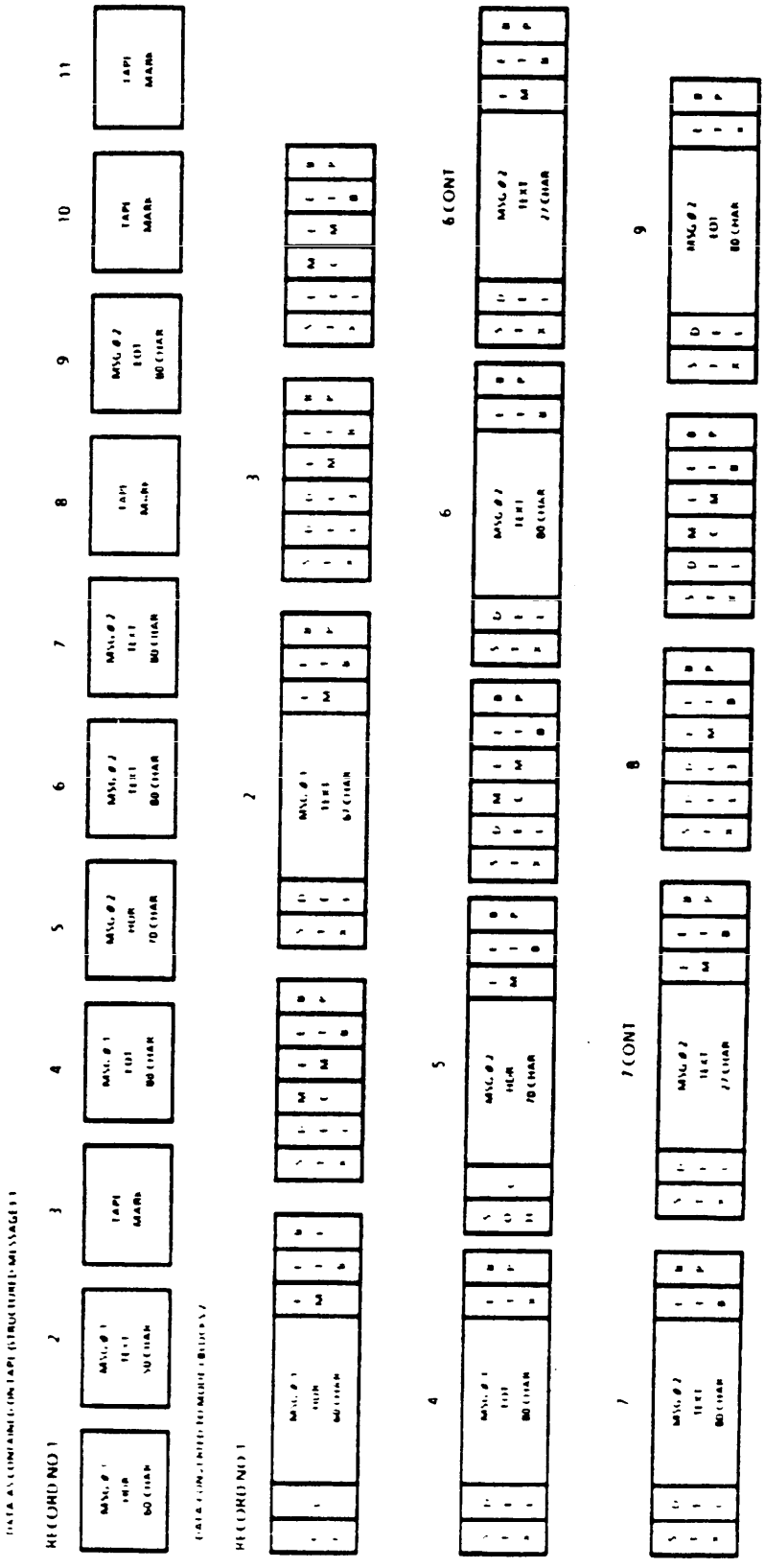


FIGURE 23. Mode I transmission nonstructured for seven-track tape format action/reaction - mode change.

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NOTES: 1/ MAGNETIC TAPE CONTAINING ONE OR MORE STRUCTURED BINARY STREAM MESSAGES WITH VARIABLE-LENGTH RECORDS. HEADER, TEXT, AND EOT FOR EACH MESSAGE ON TAPE

2/ SEE FIGURE 25 FOR TRANSMISSION OF THESE MODE I BLOCKS.

FIGURE 24. Mode I structured format for nine-track tape.

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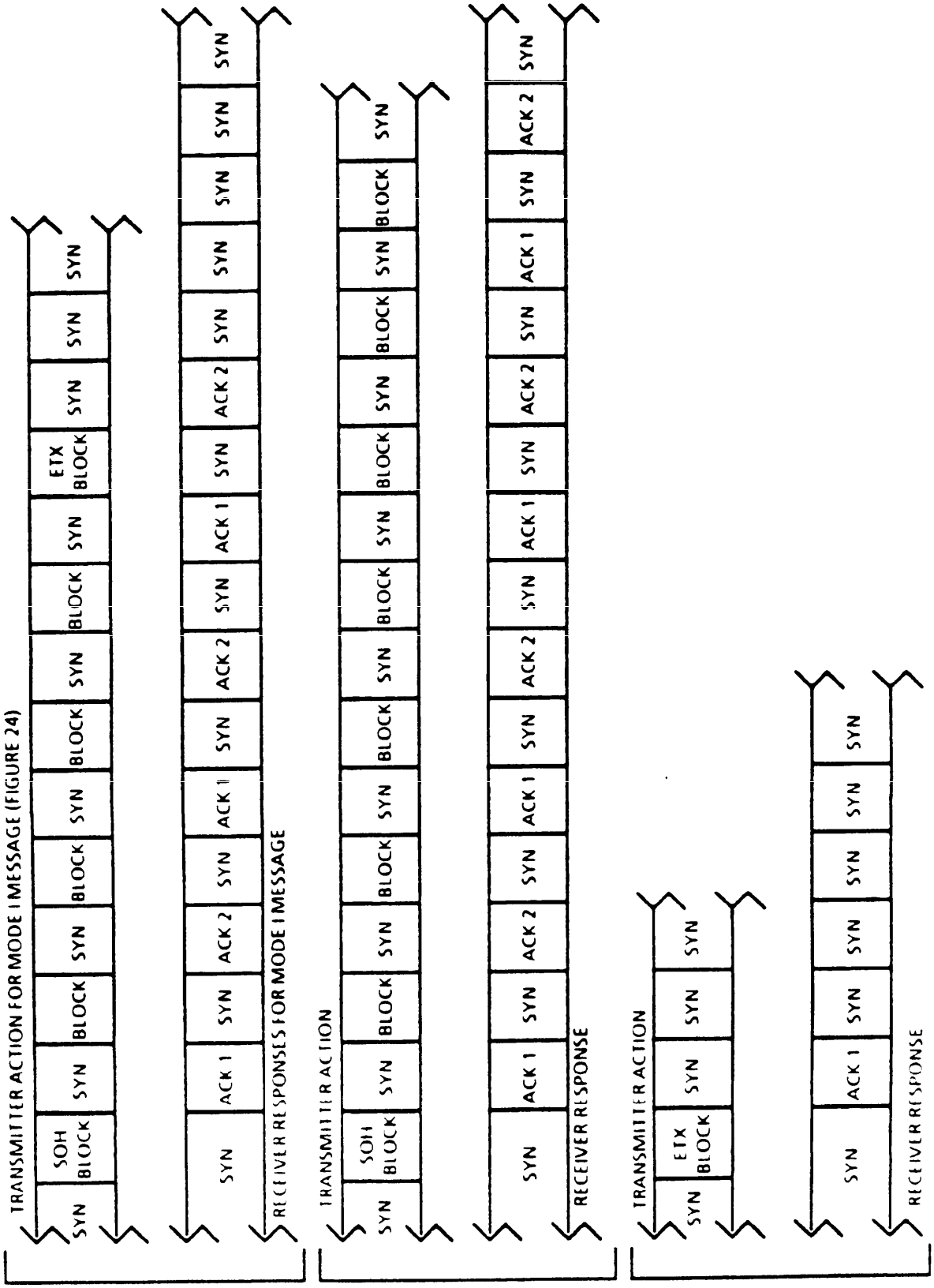
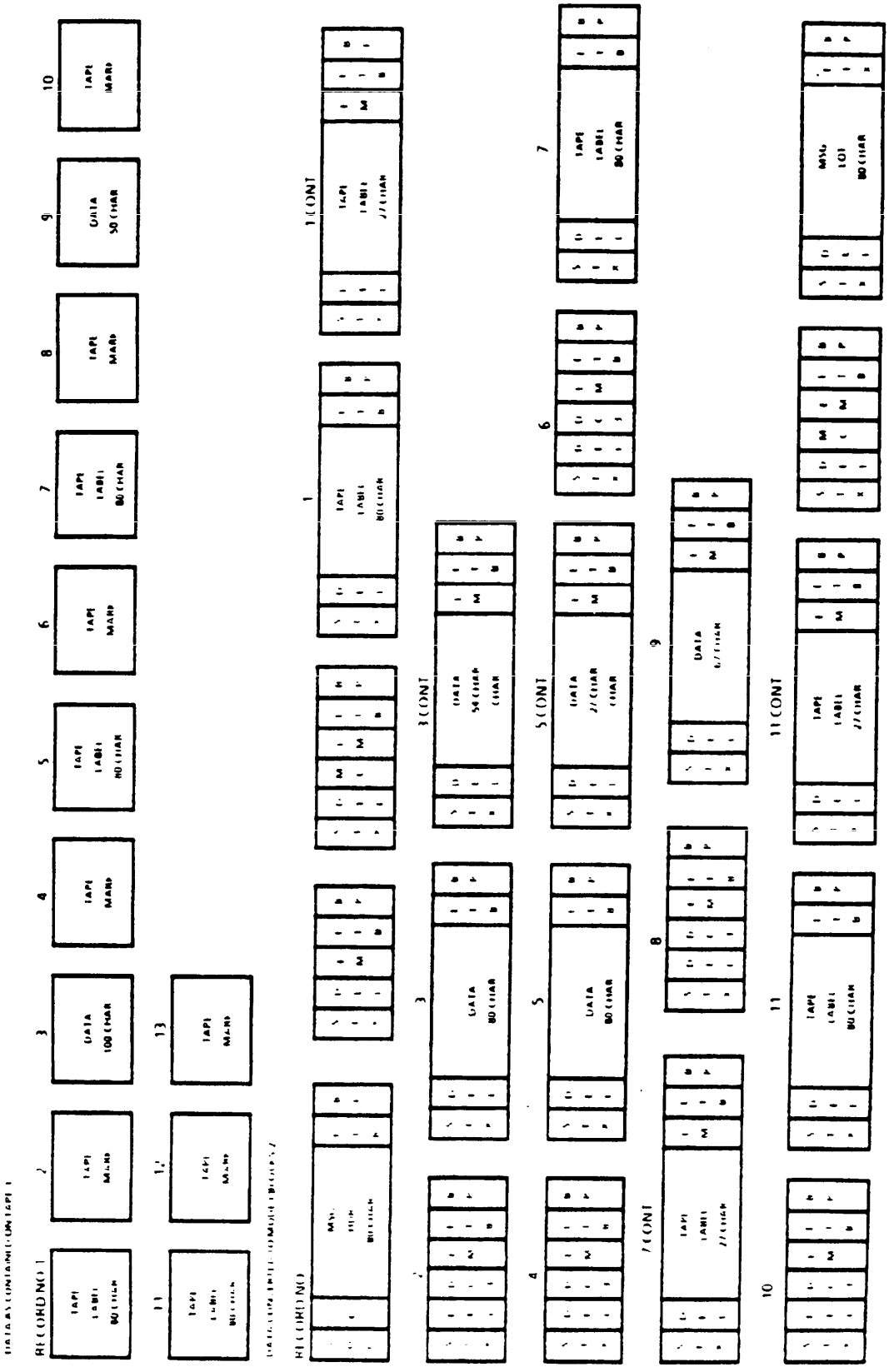


FIGURE 25. Mode I transmission structured format for nine-track tape action/reaction.

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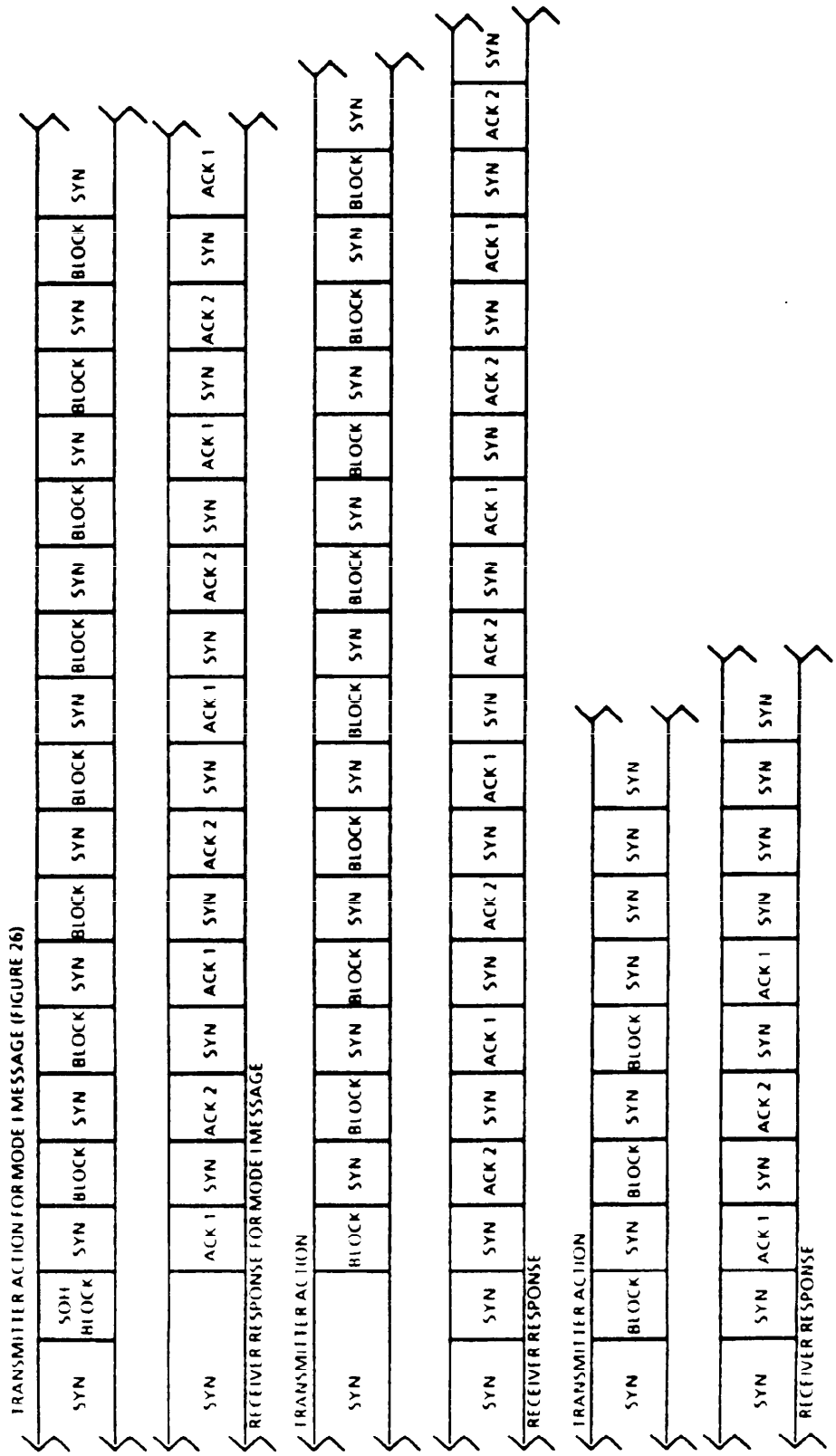
NOTES 1/ MAGNETIC TAPE CONTAINING ONLY ONE NONSTRUCTURED BINARY STREAM MESSAGE WITH VARIABLE-LENGTH RECORDS. HEADER AND EOT NOT ON TAPE.

2/ SEE FIGURE 27 FOR TRANSMISSION OF THESE MODE I BLOCKS

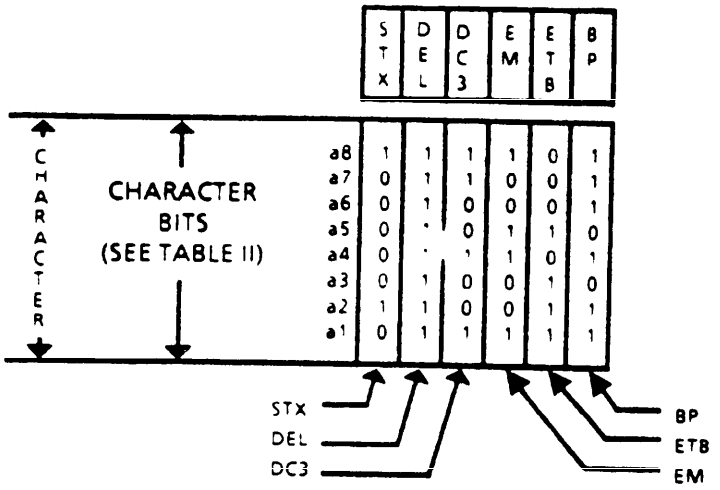
FIGURE 26. Mode I nonstructured format for nine-track tape.

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TRANSMITTER ACTION FOR MODE I MESSAGE (FIGURE 26)
SYN
SOH BLOCK
SYN
BLOCK
SYN
BLOCK
SYN

APPENDIX C



NOTE: DC3 is a Mode I data character

FIGURE 28. Mode I example of DC3 block - end of record.

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APPENDIX D

FLOW CHART (LOGICAL DESIGN) FOR MODE I TERMINAL DESIGN

This appendix contains general information in support of MIL-STD-188-171. Appendix D is a nonmandatory part of this document.

10. GENERAL

10.1 Scope. This appendix contains the flow charts for a Mode I terminal design. The implementation of this appendix is not mandatory.

10.2 Application. This appendix may be used as a guide in developing Mode I channel coordination procedures.

20. REFERENCE DOCUMENTS. Not applicable.

30. DEFINITIONS. Not applicable.

40. GENERAL REQUIREMENTS

40.1 Minimum criteria. Any system or terminal implementing Mode I shall conform to the requirements of MIL-STD-188-171.

50. SPECIFIC

50.1 Requirements. State diagrams and flow charts in this appendix (figures 29 through 43) are examples that have been used in successful implementations of Mode I channel coordination procedures.

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immediately transmission of the next block at a character boundary and

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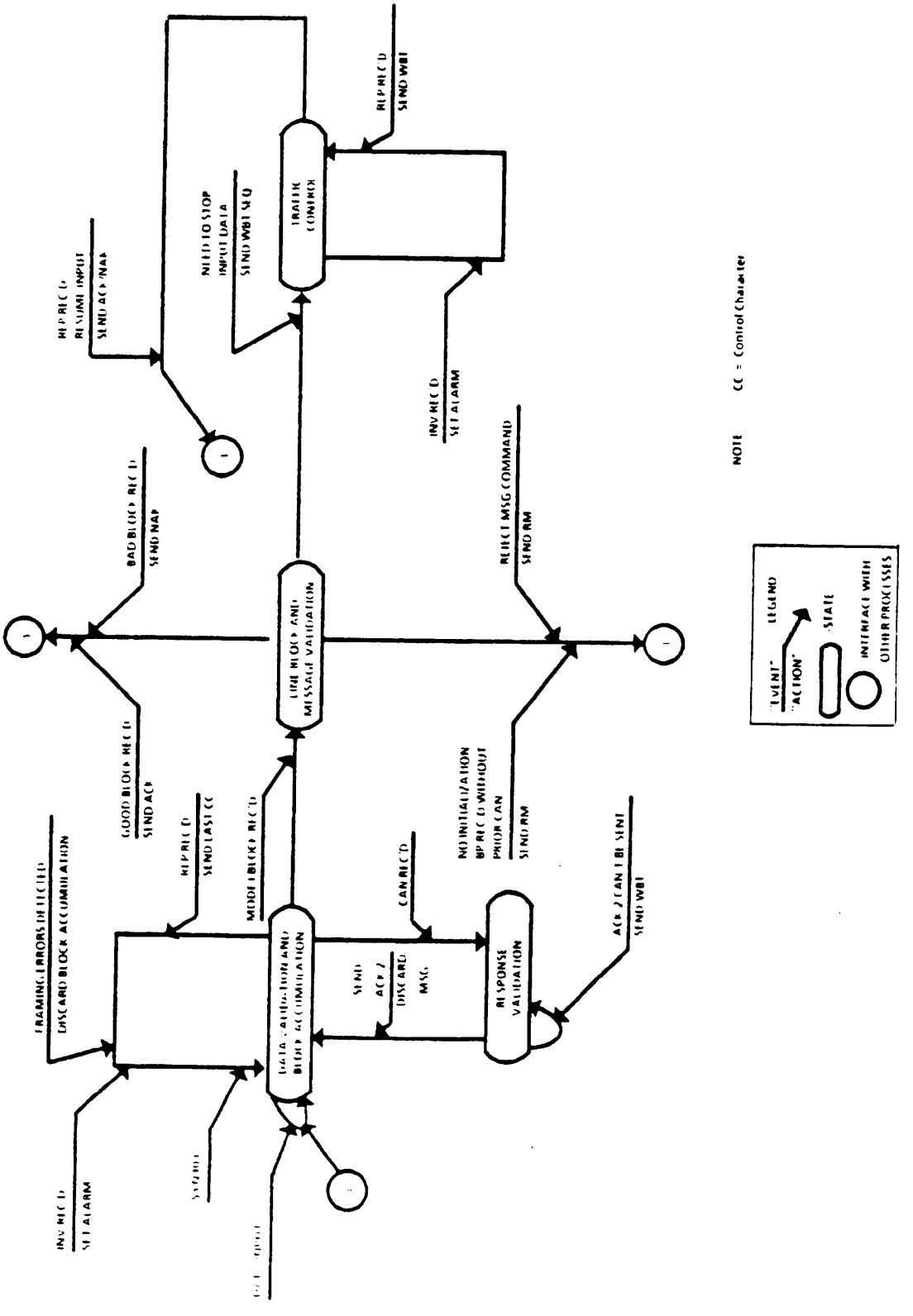
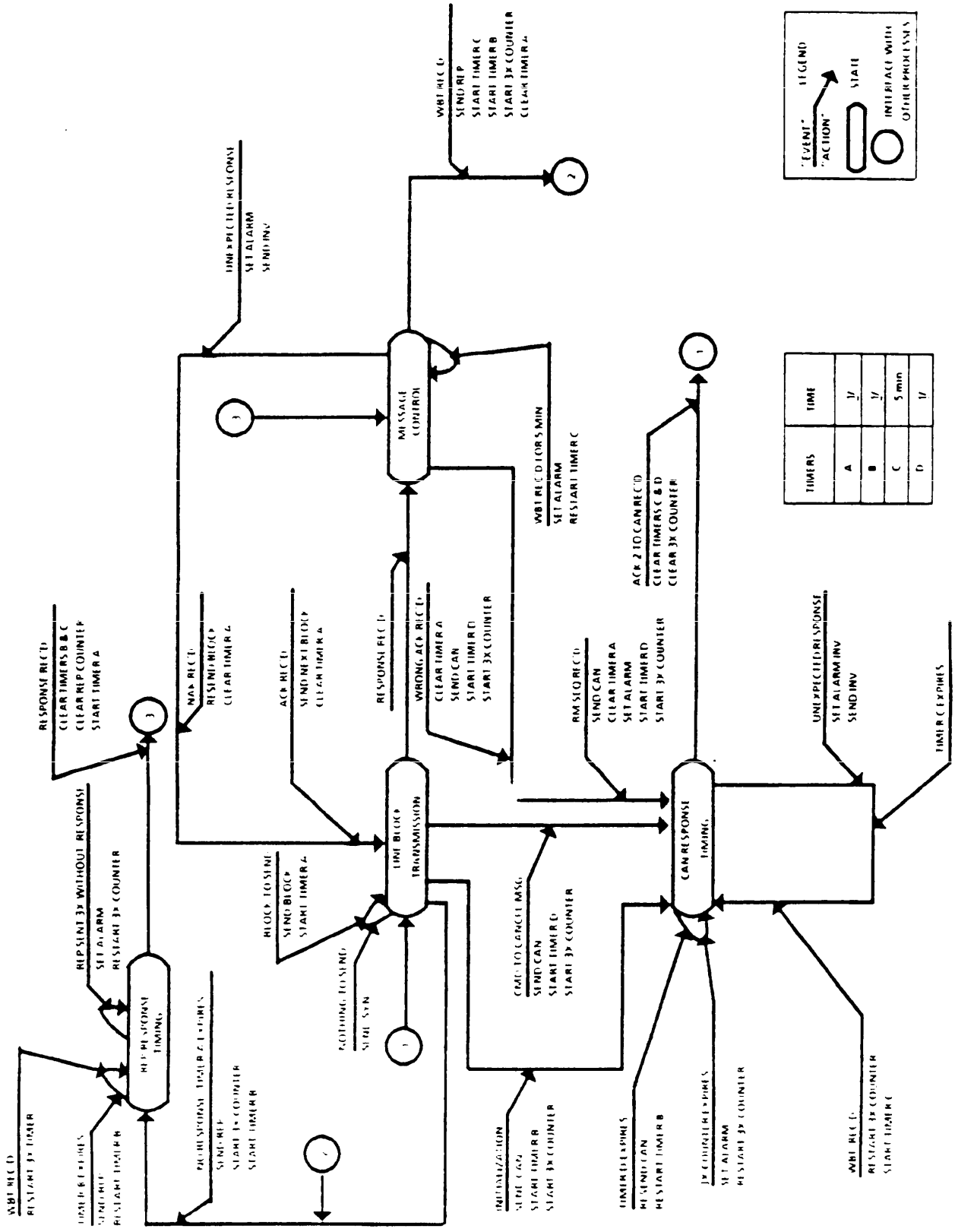


FIGURE 29. Mode I message reception.

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NOTE: 1/ Transmitter Answer Timer (See ¶5.9.1.1)

FIGURE 30. Mode I message transmission.

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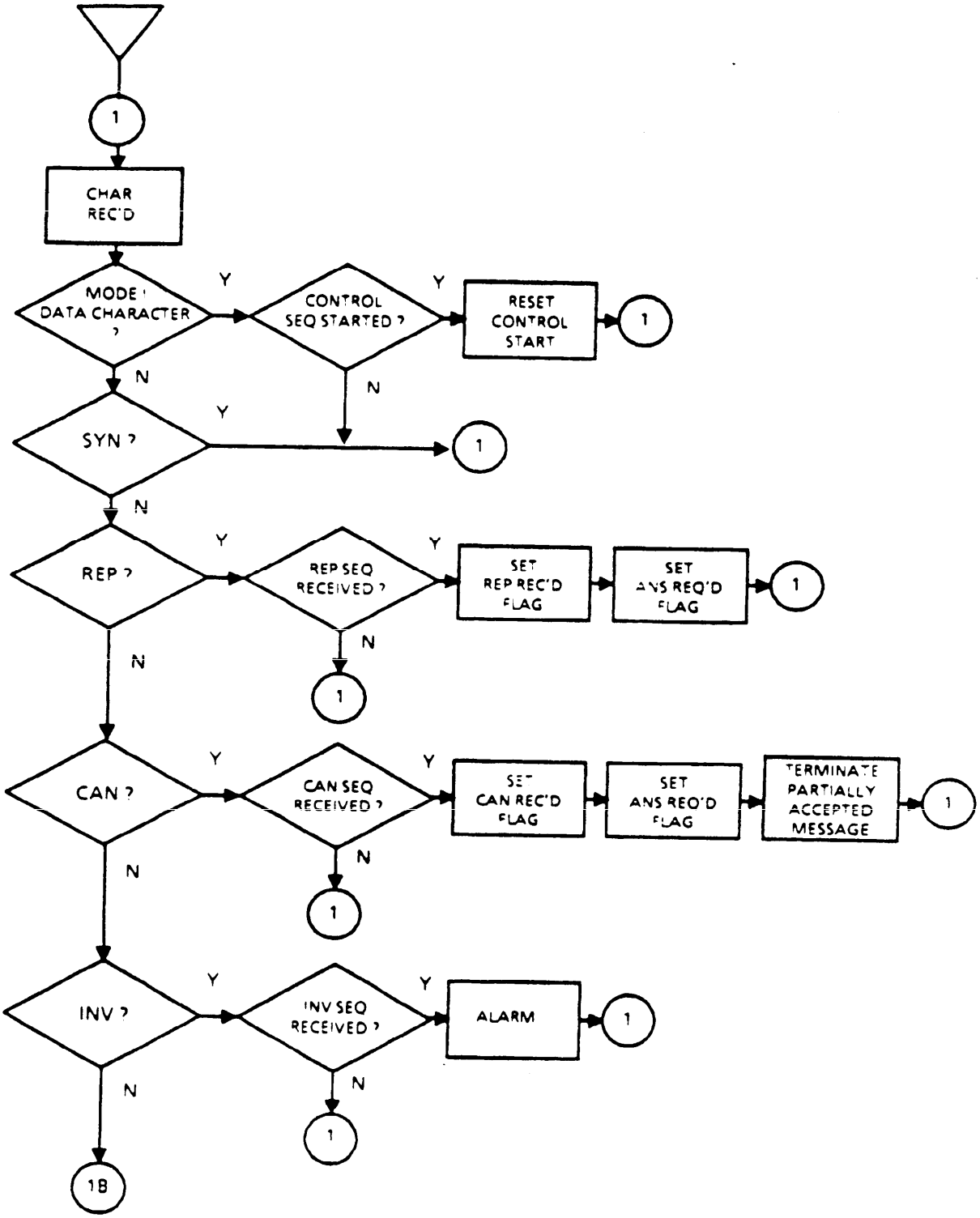


FIGURE 31. Mode I receive start of block position.

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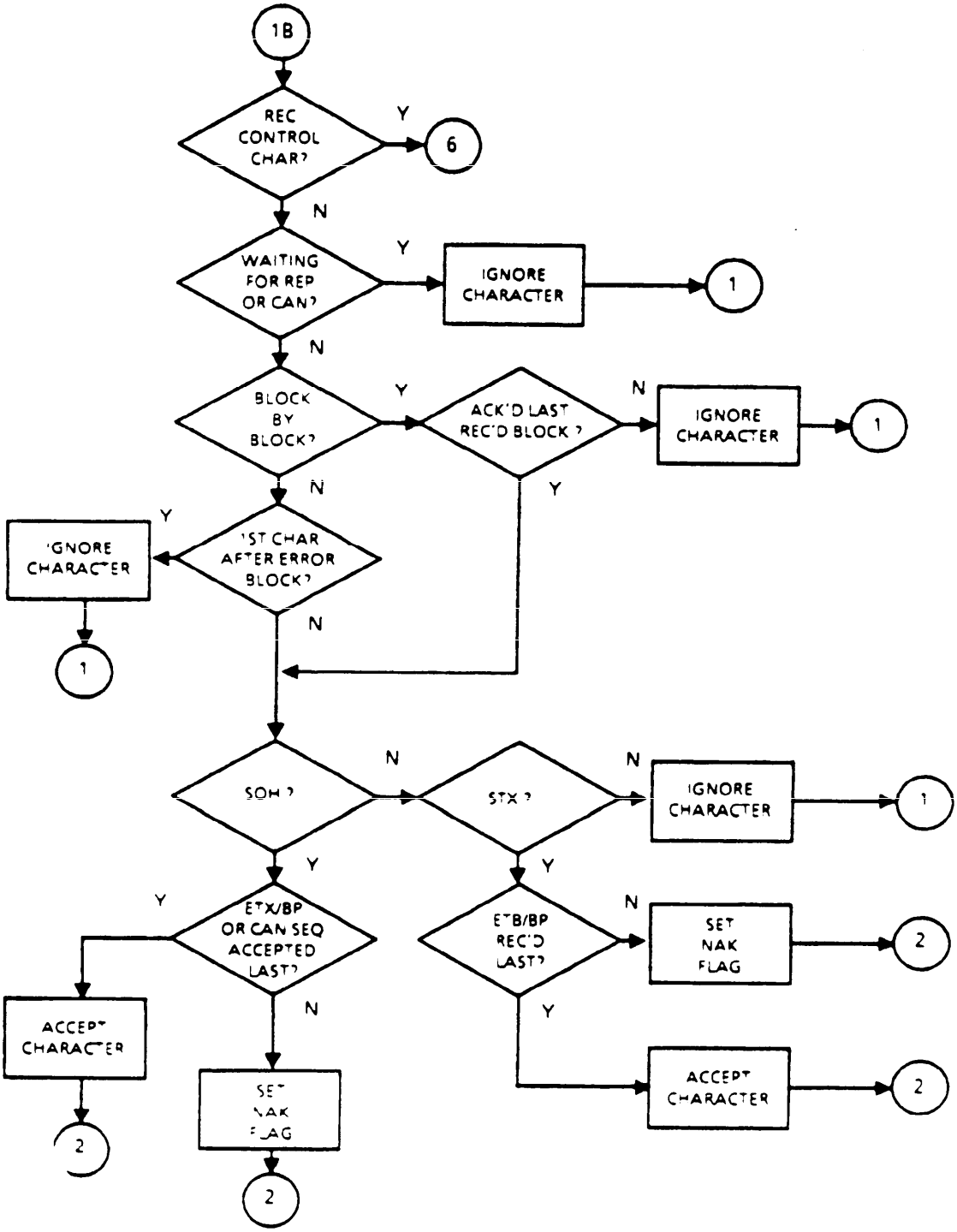


FIGURE 32. Mode I continuation of receive start of block position.

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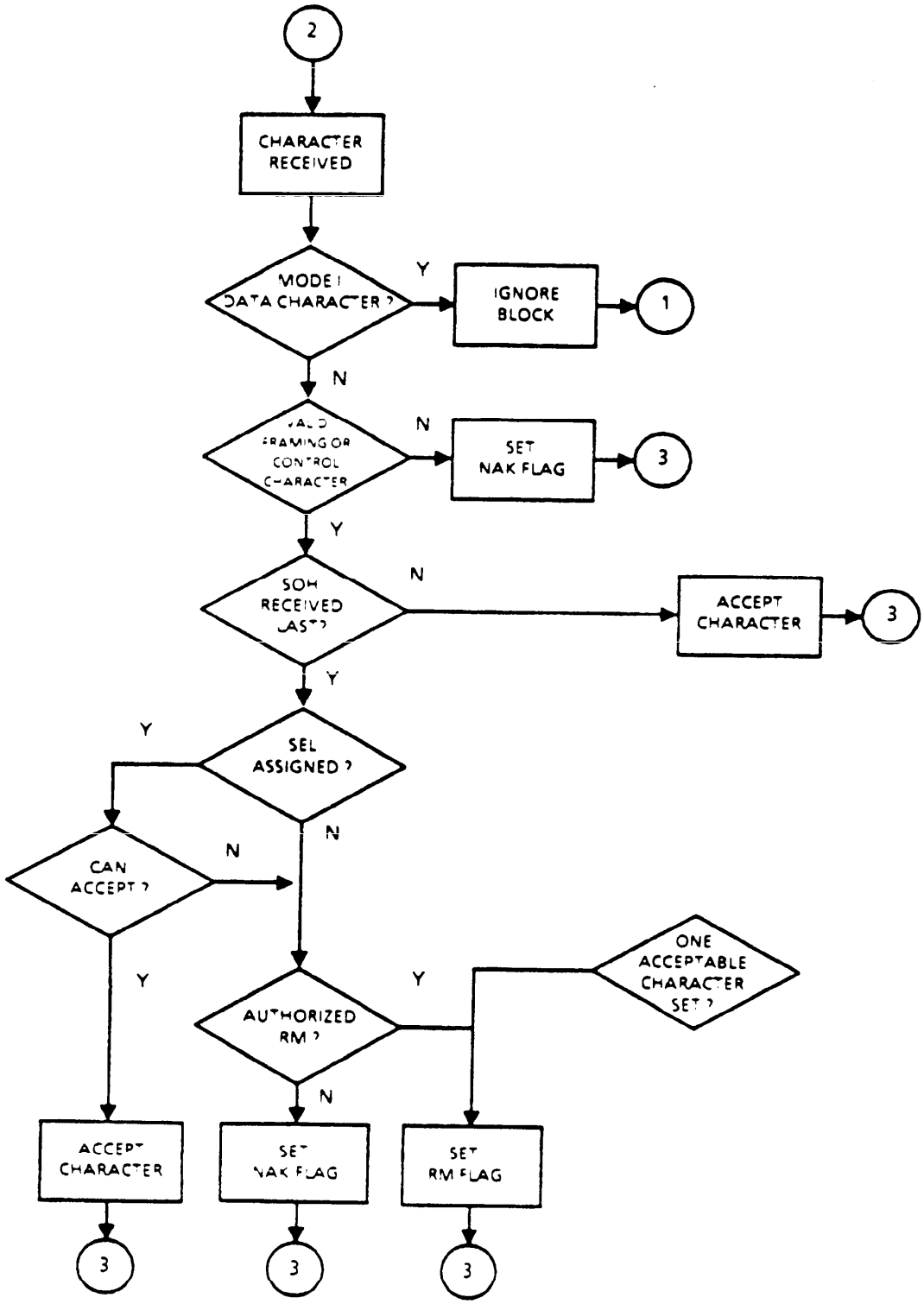


FIGURE 33. Mode I receive second framing position.

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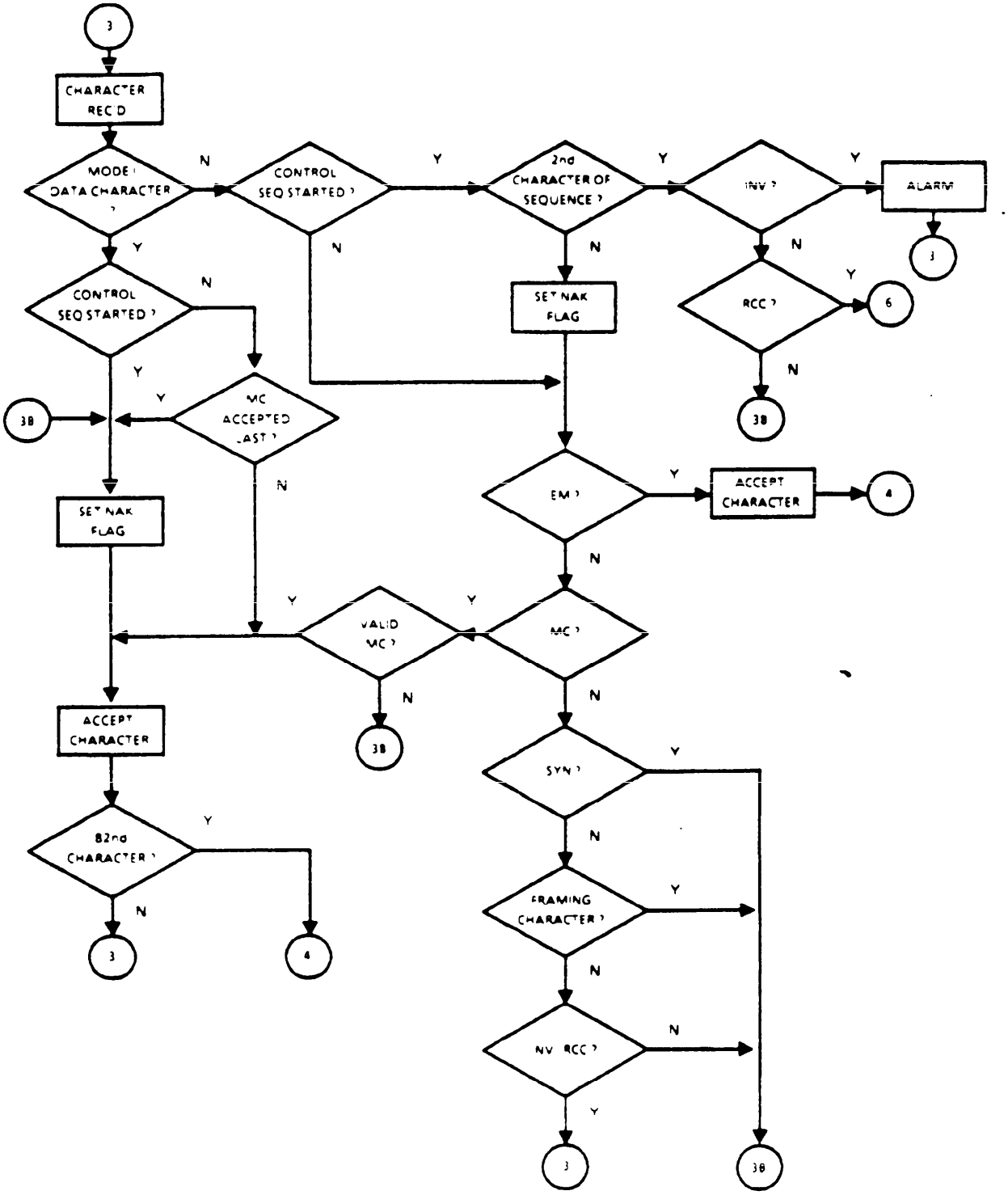


FIGURE 34. Mode I receive data position.

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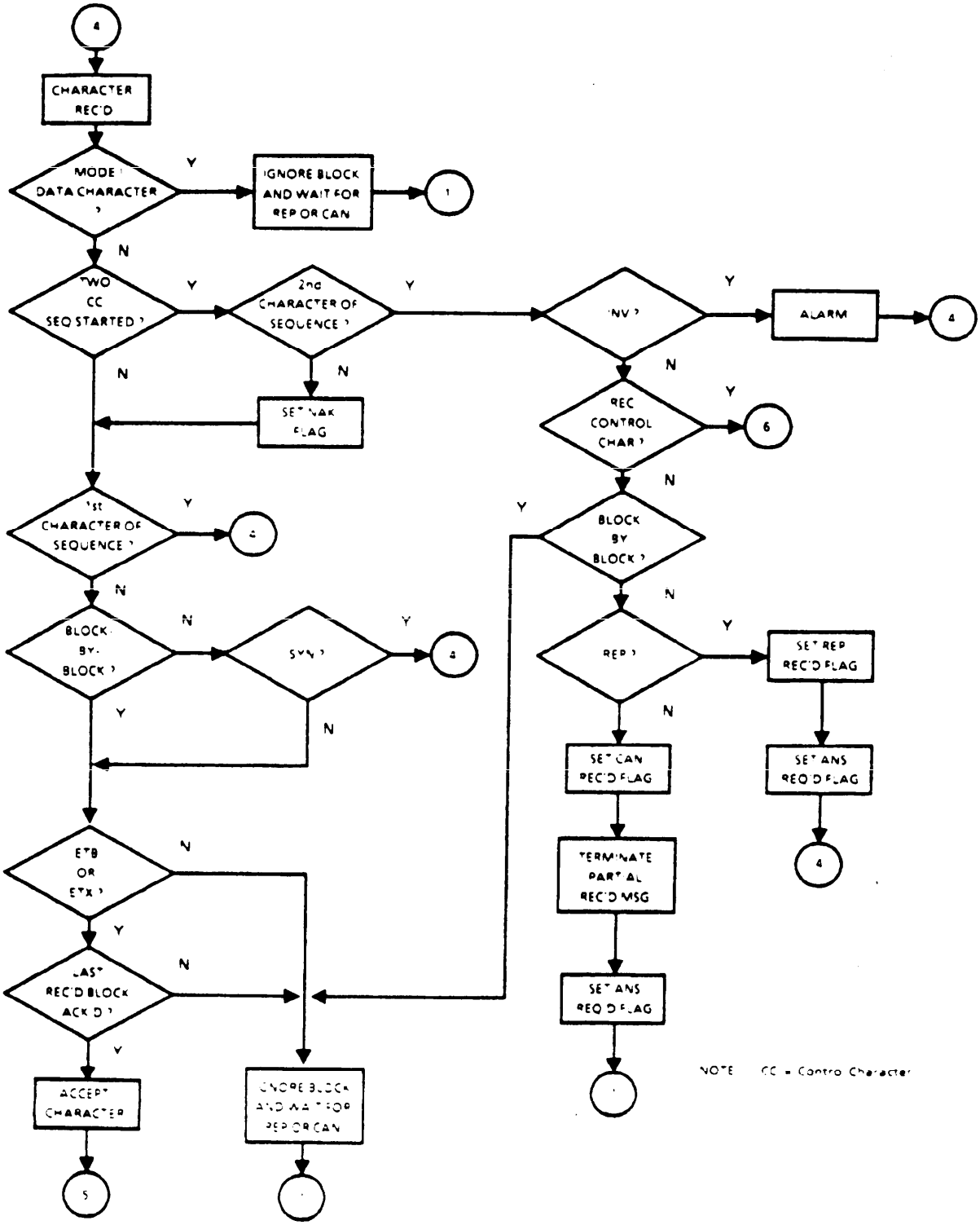


FIGURE 35. Mode I receive end of block framing position.

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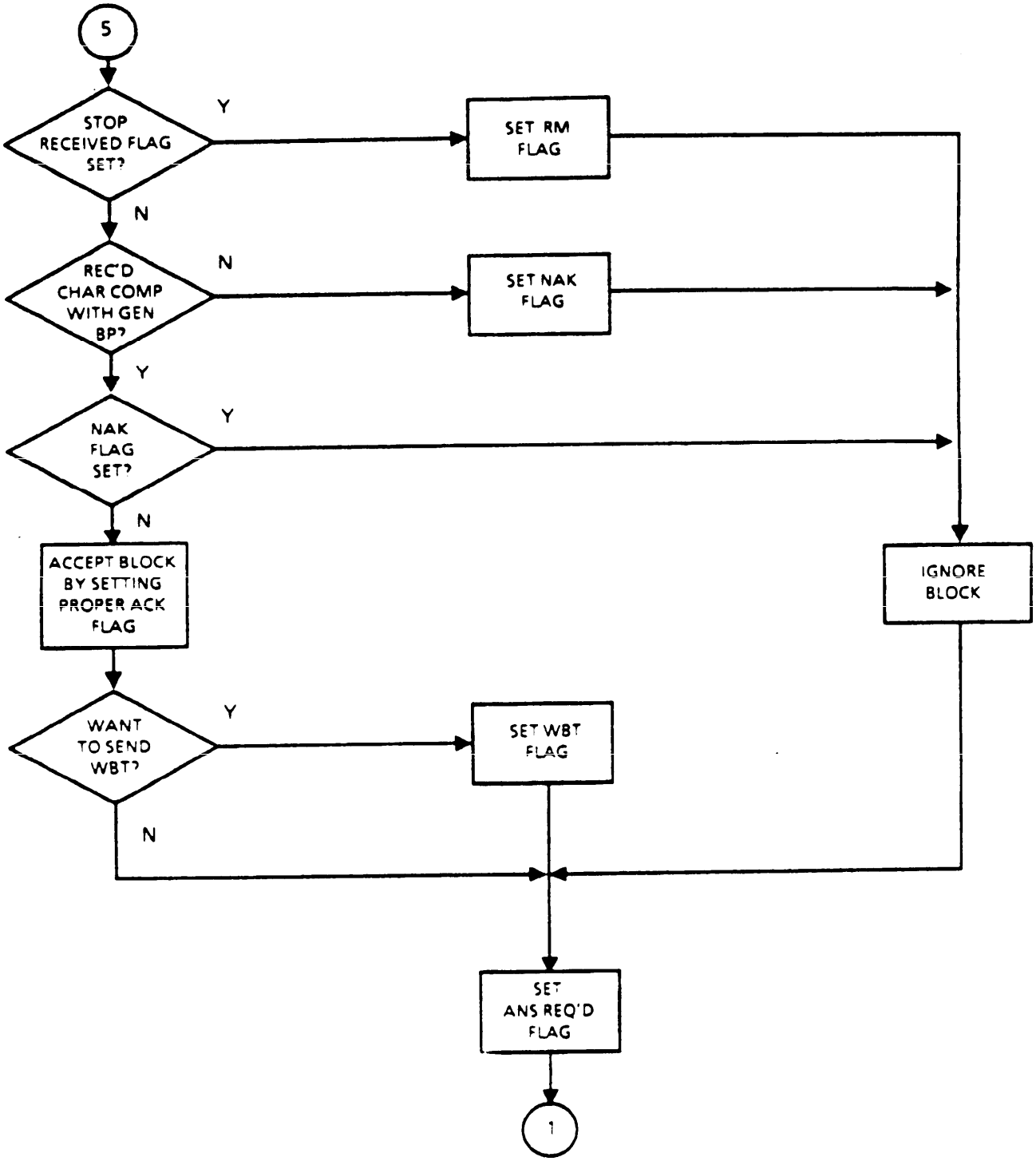


FIGURE 36. Mode I receive block parity (BP) position.

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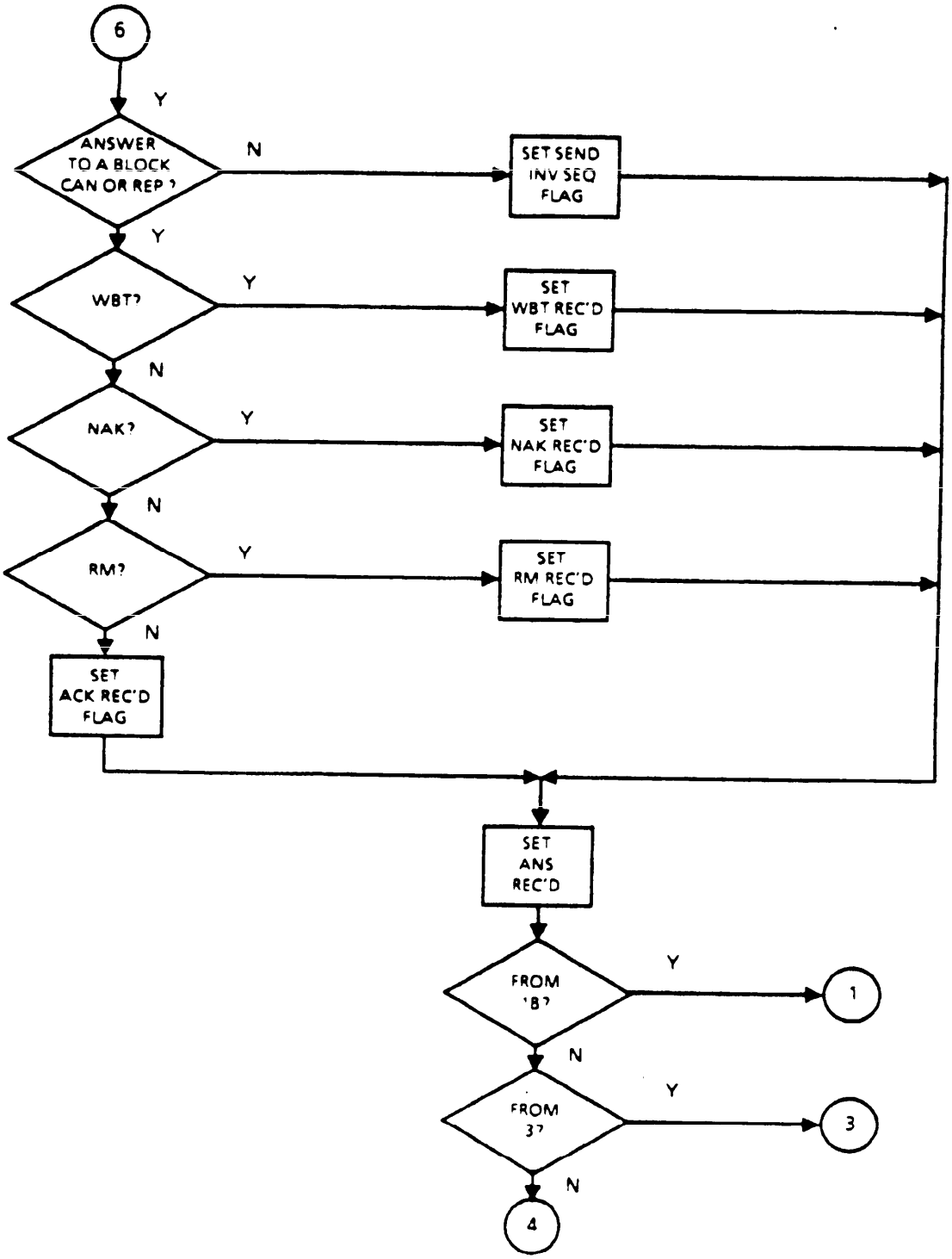


FIGURE 37. Mode I receive control sequence.

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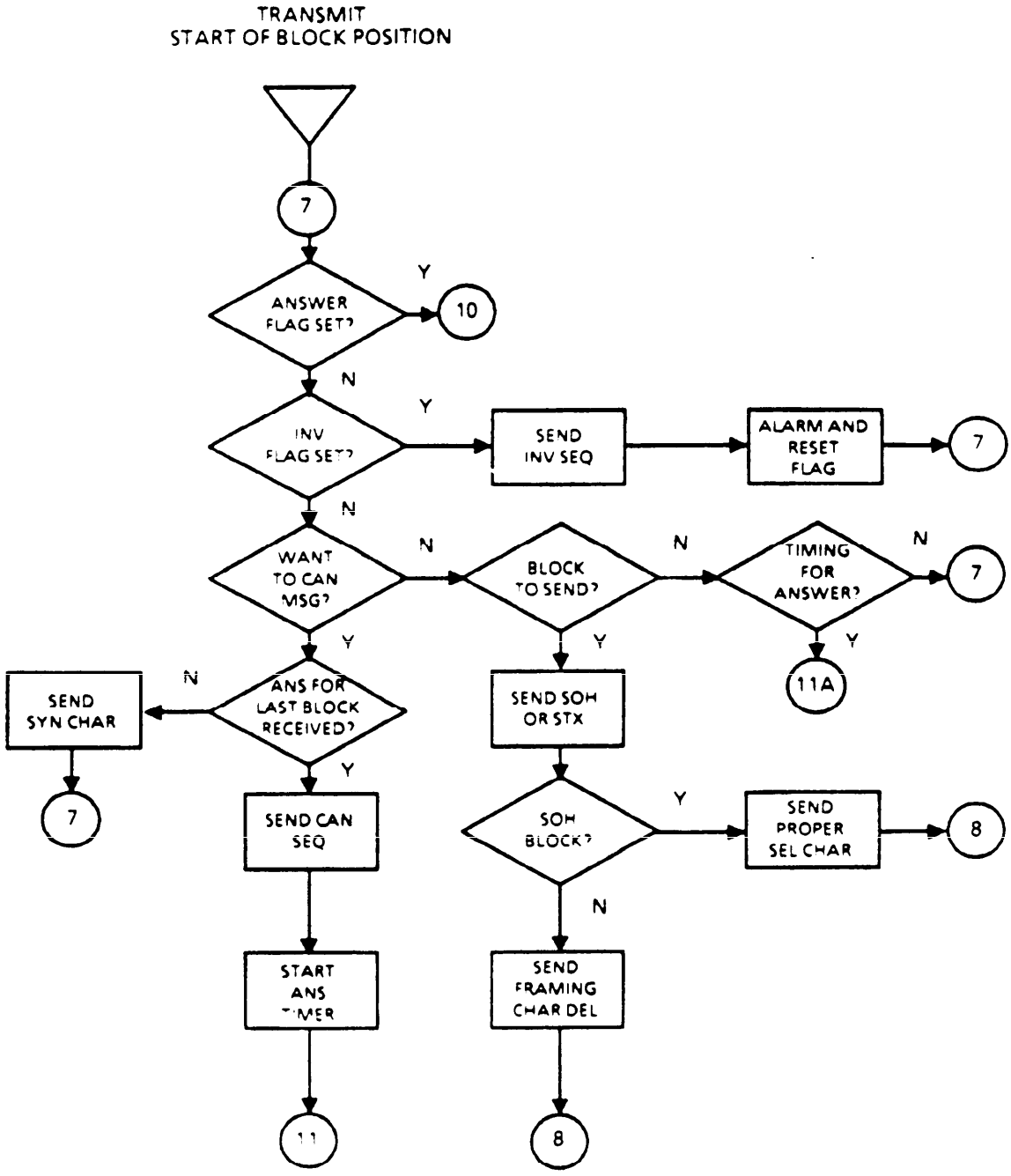


FIGURE 38. Mode I transmit start of block position and second framing position.

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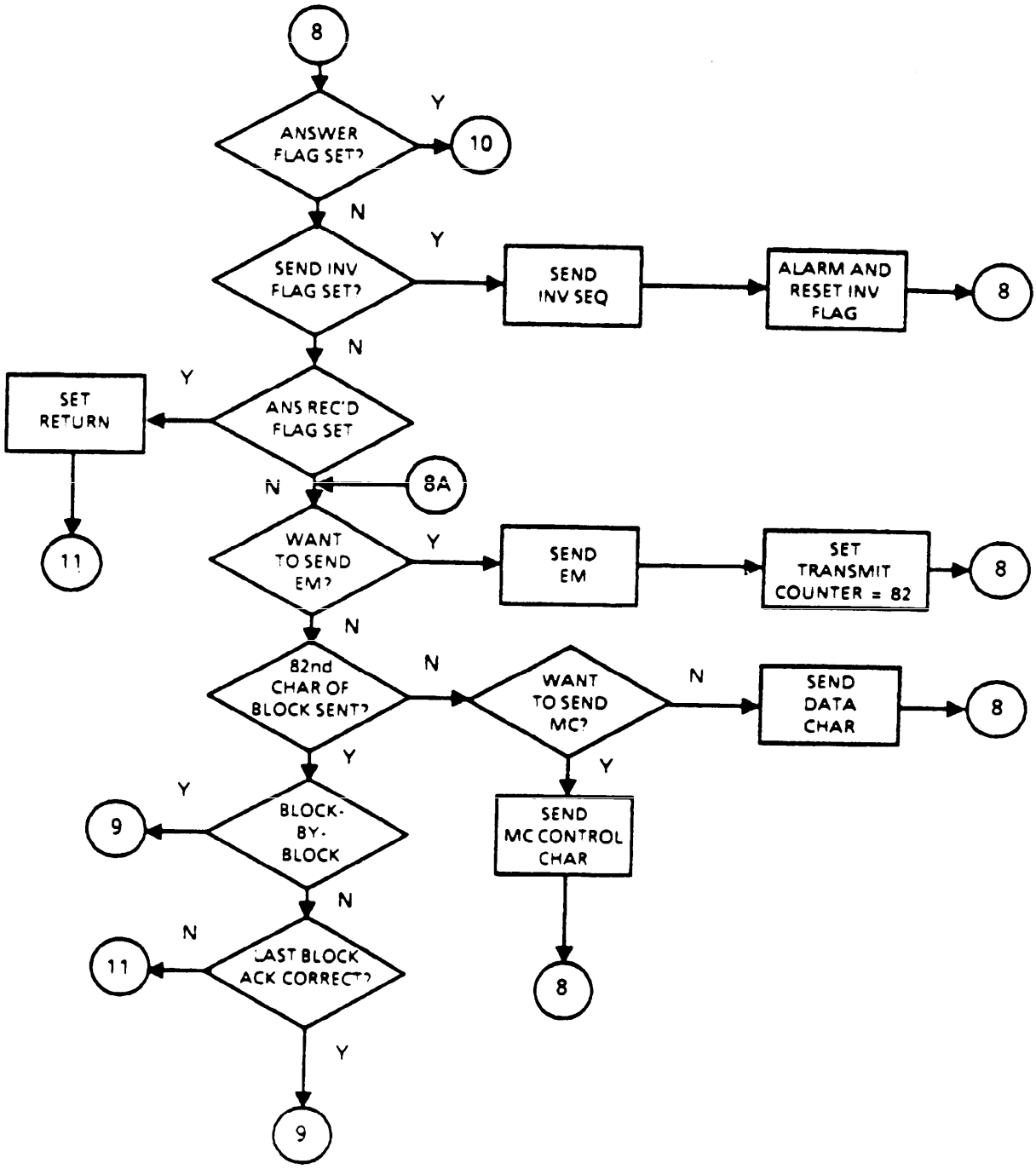


FIGURE 39. Mode I transmit data position.

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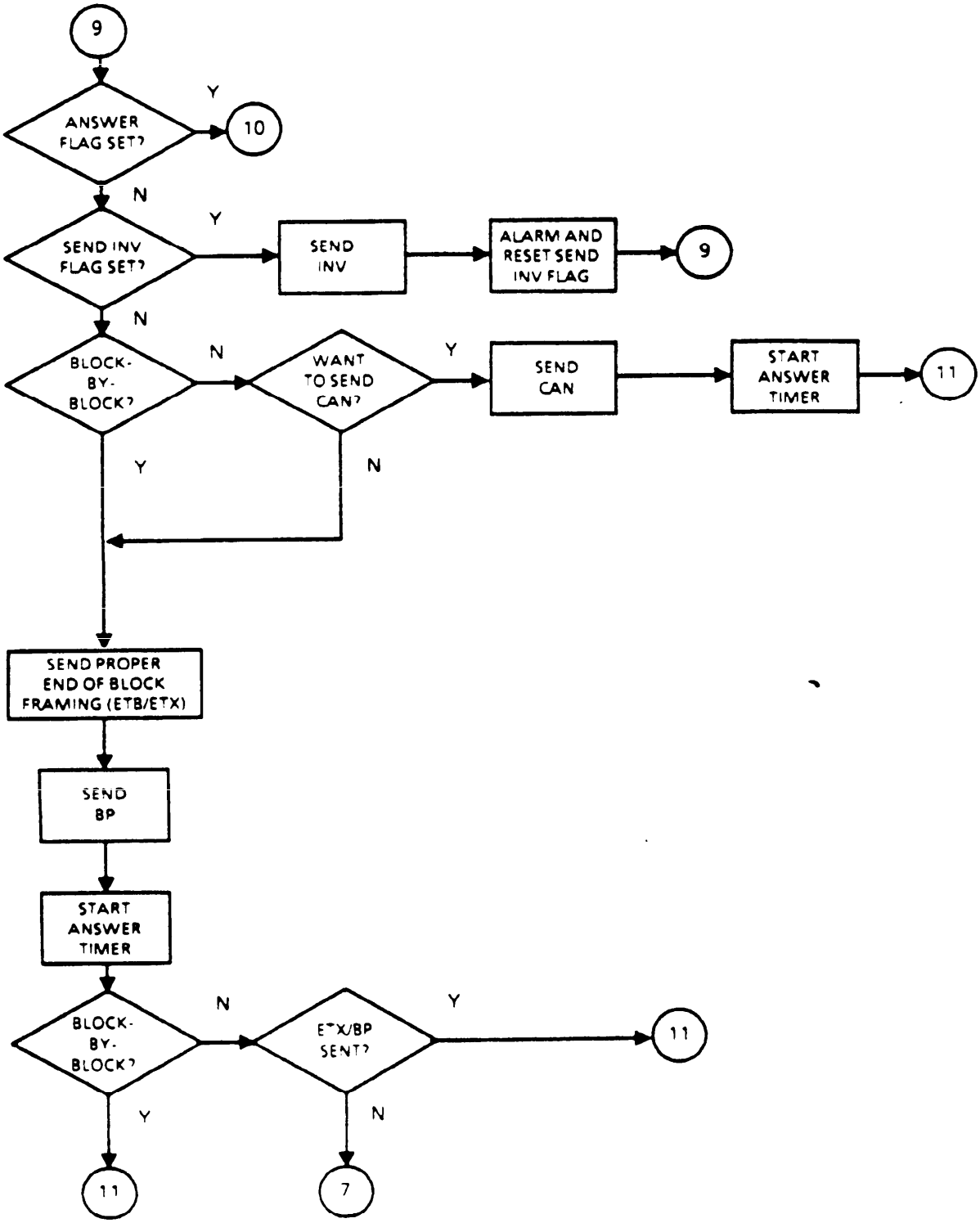


FIGURE 40. Mode I transmit end of block position.

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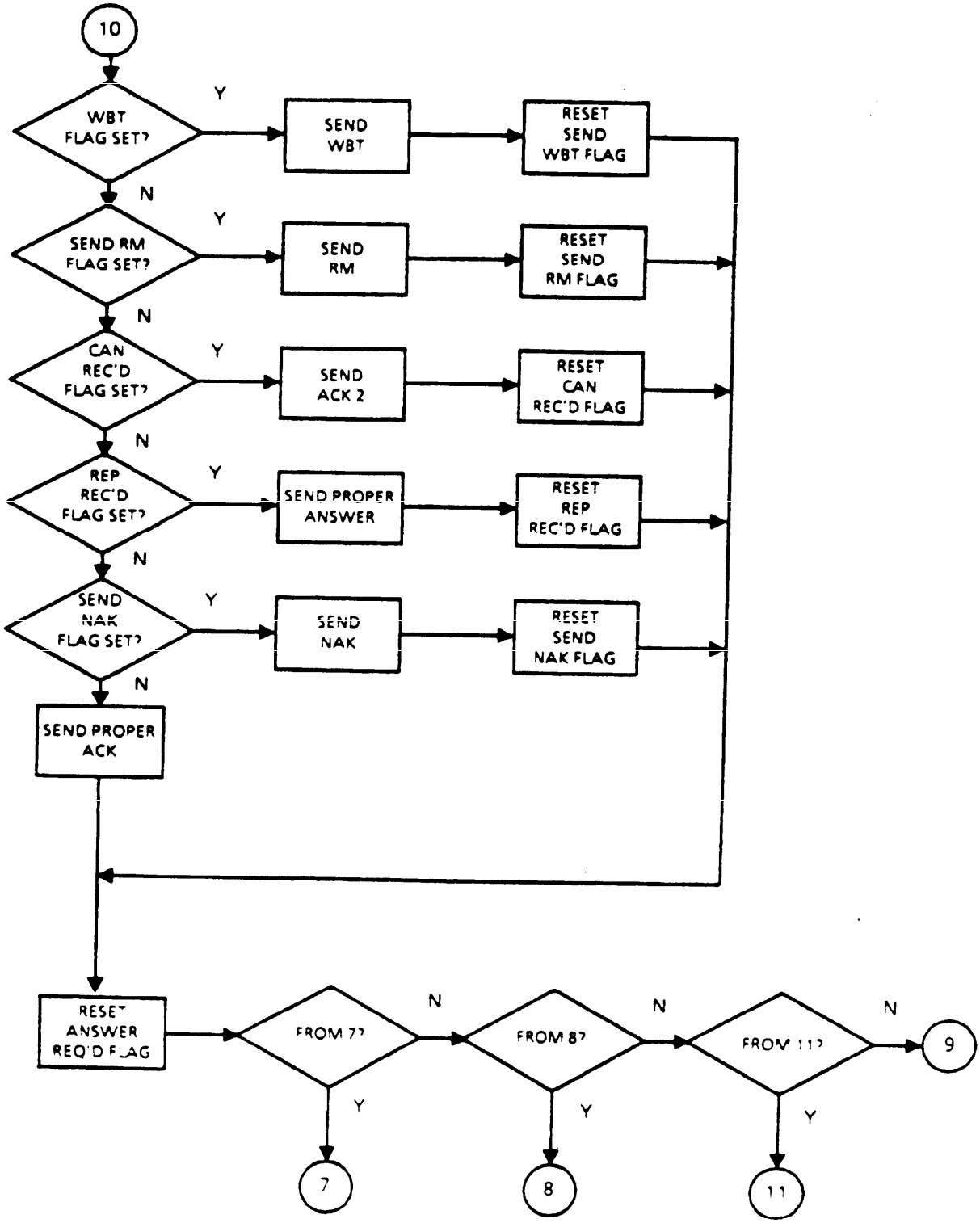


FIGURE 41. Mode I transmit answer position.

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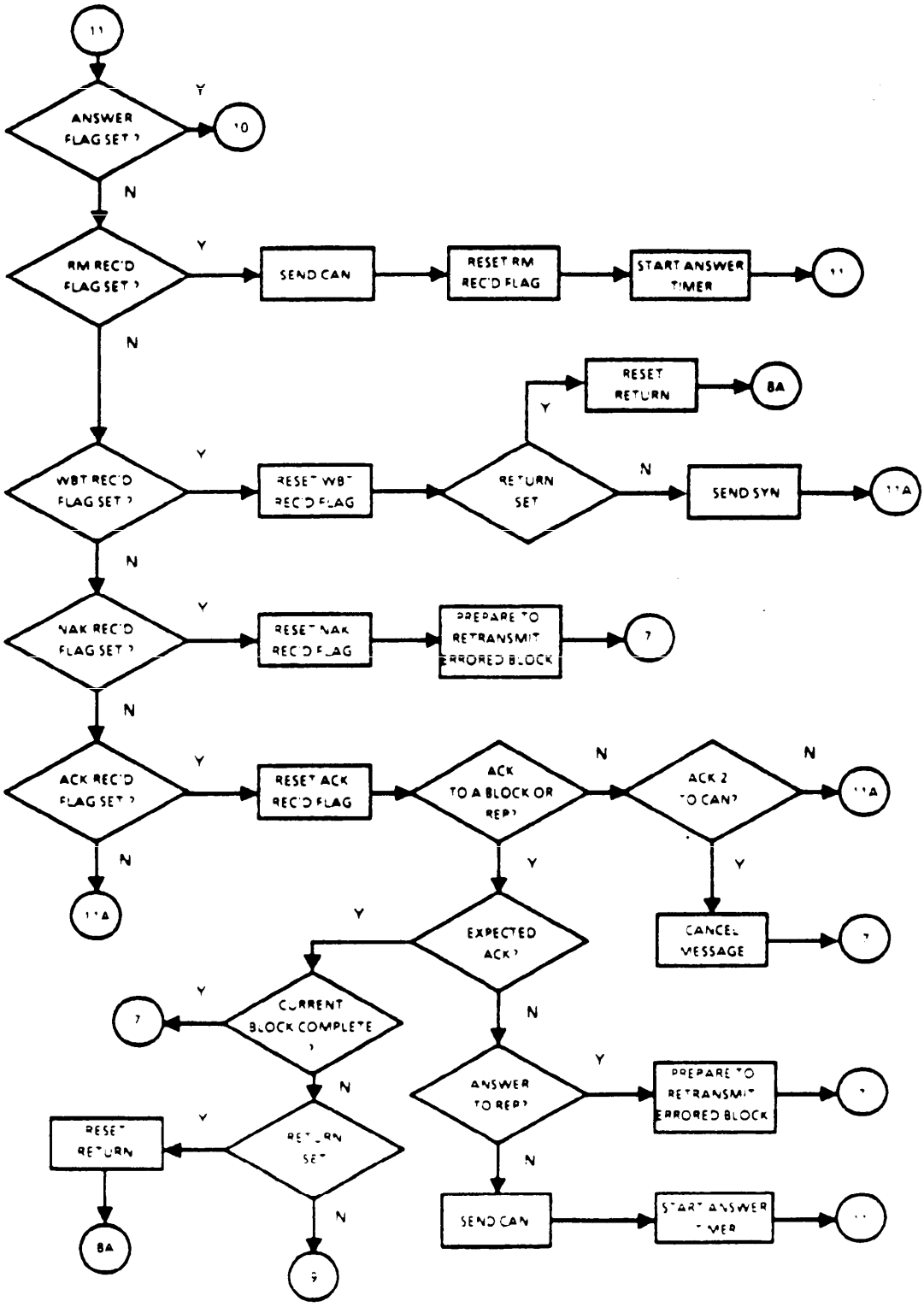


FIGURE 42. Mode I transmit control.

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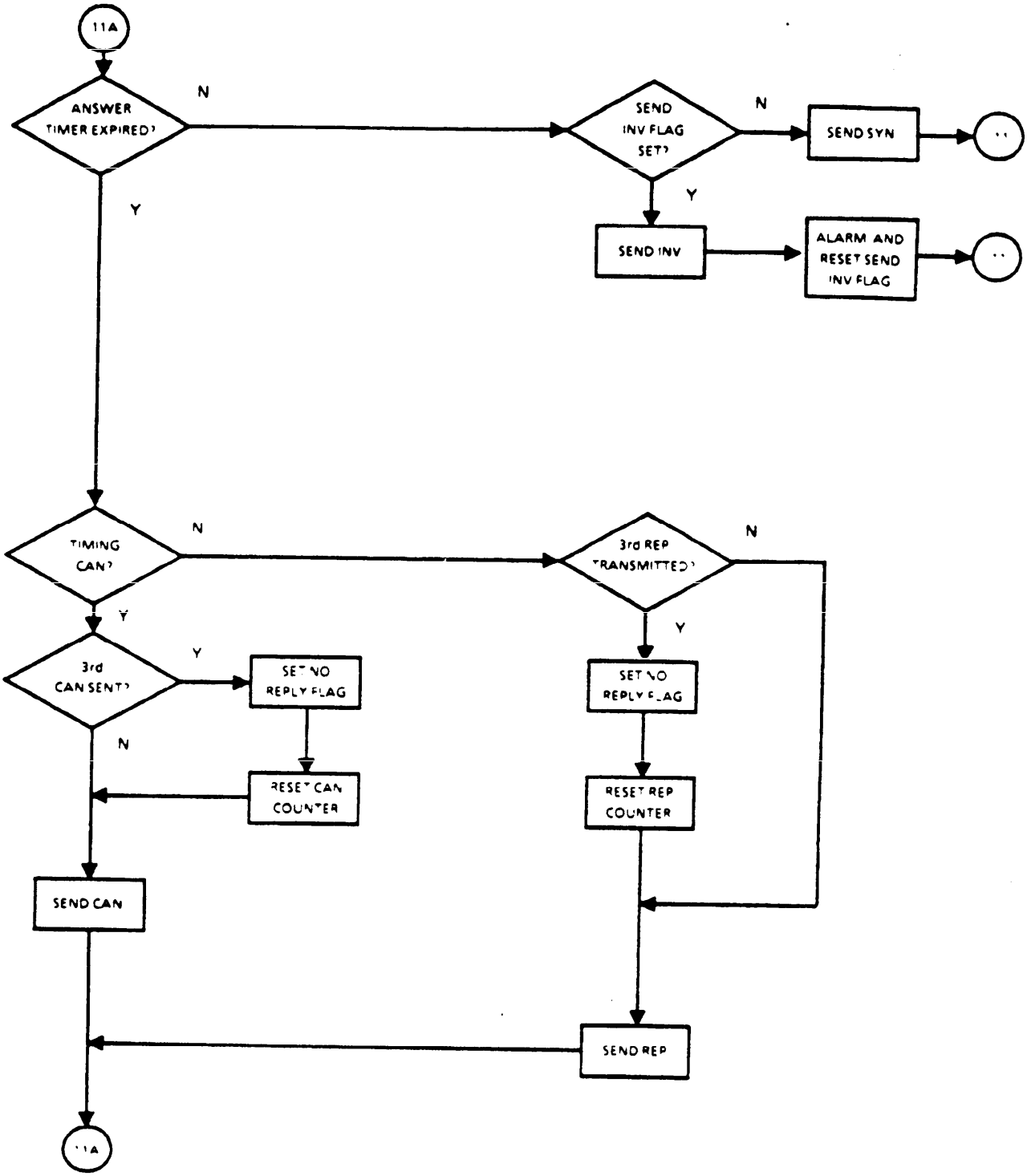


FIGURE 43. Mode I continuation of transmit control.

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